

WM9PD – Network Security

DesX Zolutions Ltd. Network Design and Implementation

No – 5750945

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Introduction

DesX Zolutions is a growing architectural design firm who has 5 departments with 160 employees. The network is created on Cisco Packet tracer with a combination of switches, routers, PC, wireless access points, laptops and server. It is a working network that supports inter-communication, DHCP, AAA, server access and more. The designing, implementing and application of security policies are also configured. This report documents the overall implementation, outcome and justification of this network design.

Task 1: Network Design and Planning

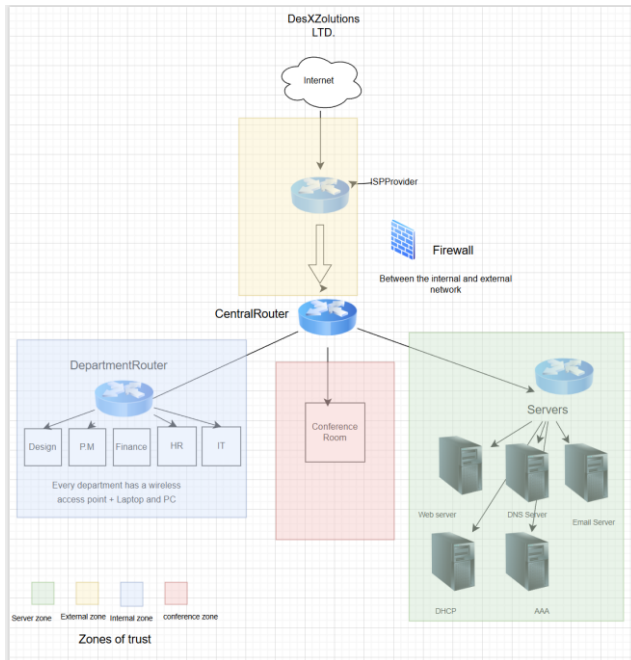


Fig 1 – network design

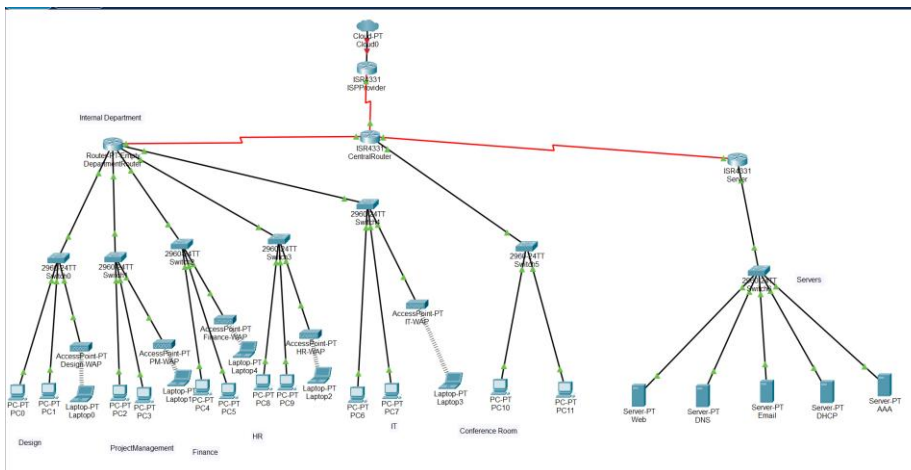


Fig 2 – network in packet tracer

The network design follows a star topology in a hierarchical structure which makes it easy to implement and visualise the network of DesX Solutions Ltd. This ensures any user can understand how the network operates. A combination of routers, pc, laptops, wireless access points, servers and switches are used in this network. The ISR4331 is used in the central router and server router as the securityk9 module is preinstalled. This makes it easier for ACL and ZBF configuration. For the department router, the router pt-empty is used due to the large availability of giga ethernet ports. The segmentation of this network is done based on the zones of trust, dividing the network on zones of internal, DMZ, conference and external. The internal consists of the entire business departments making it a very important part of the network. The conference room has a mid-level trust since it is accessible and used by everyone. The conference room is connected directly to the central router. On the other hand, the DMZ zone which is a critical part of the network is placed under the server router, which consists of Web, DNS, Email, DHCP and AAA. The external network is simulated with a router named ISP-Provider router. The network consists of 3 routers since DesX Solutions Ltd is still a growing company and large number of routers make the network expensive.

Department	Hosts	Network Address	Subnet Mask	Prefix	IP address range	Broadcast address	Default Gateway
Design	55	192.168.10.0	255.255.255.192	/26	192.168.10.1-192.168.10.62	192.168.10.63	192.168.10.1
Project Management	40	192.168.10.64	255.255.255.192	/26	192.168.10.65-192.168.10.126	192.168.10.127	192.168.10.65
Finance	30	192.168.10.128	255.255.255.224	/27	192.168.10.129-192.168.10.158	192.168.10.159	192.168.10.129
HR	25	192.168.10.160	255.255.255.224	/27	192.168.10.161-192.168.10.190	192.168.10.191	192.168.10.161
IT	10	192.168.10.192	255.255.255.240	/28	192.168.10.193-192.168.10.206	192.168.10.207	192.168.10.193
Servers	Servers – 5 servers	192.168.10.208	255.255.255.240	/28	192.168.10.209-192.168.10.222	192.168.10.223	192.168.10.209
Conference Room	Conf room- 2-5 pcs	192.168.10.224	255.255.255.240	/28	192.168.10.225-192.168.10.238	192.168.10.239	192.168.10.225
Depart Central	- 2	192.168.10.240	255.255.255.252	/30	192.168.10.241-192.168.10.242	192.168.10.243	N/A
Central Server	- 2	192.168.10.244	255.255.255.252	/30	192.168.10.245-192.168.10.246	192.168.10.247	N/A

Table 1- IP addressing using 192.168.10.0/24

The IP VSLM addressing as seen in Table 1 is done based on the required number of hosts from the initial address of 192.168.10.0 /24. The last address of the IP is part of the broadcast address.

1. For the Design Department – 192.168.10.0 /26, there are 62 available hosts
2. For the Project Management Department– 192.168.10.64/26 there are 62 available hosts
3. For the Finance Department – 192.168.10.128/27, there are 30 available hosts
4. For the HR Department- 192.168.10.160/27, there are 30 hosts available
5. For the IT Department – 192.168.10.192/28, there are 14 available hosts
6. For the Servers – 192.168.10.208 /28

For the conference room – 192.168.10.224/28 was allocated. A /28 is a quite a large block for a conference room despite only 2-5 devices are needed. A larger subnet is chosen with the idea to allow future scalability by adding wireless devices, conferencing systems and more without having the need to re-design the network.

Task 2: Network Implementation and Design

The IP addressing table serves as a reference when adding the IP addresses onto the routers. For the routers, the first step is to add a hostname for easier implementation. A description is also added in every interface for ease. The IP addressing of the serials can be seen below.

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname CentralRouter
CentralRouter(config)#interface s
CentralRouter(config)#interface serial0/1/0
CentralRouter(config-if)#description ToDepartmentRouter
CentralRouter(config-if)#ip address 192.168.10.242 255.255.255.252
CentralRouter(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
CentralRouter(config-if)#exit
CentralRouter(config)#interface serial0/2/0
CentralRouter(config-if)#description ToServerRouter
CentralRouter(config-if)#ip address 192.168.10.245 255.255.255.252
CentralRouter(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/2/0, changed state to down
CentralRouter(config-if)#
```

Fig 1 – Ip addressing in central router between Depart and Server Router

The conference room is directly linked to the central router. The ip addressing can be seen as follows.

```
CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#interface gigabitEthernet0/0/0
CentralRouter(config-if)#description ToConferenceRoom
CentralRouter(config-if)#ip address 192.168.10.225 255.255.255.240
CentralRouter(config-if)#no shutdown
CentralRouter(config-if)#exit
CentralRouter(config)#exit
CentralRouter#
```

Outcome

The ip addressing on the central router can be seen with the 'show ip interface brief' command

```
CentralRouter#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  192.168.10.225  YES manual  up          up
GigabitEthernet0/0/1  unassigned      YES unset   administratively down down
GigabitEthernet0/0/2  unassigned      YES unset   administratively down down
Serial0/1/0        192.168.10.242  YES manual  up          up
Serial0/1/1        unassigned      YES unset   administratively down down
Serial0/2/0        192.168.10.245  YES manual  up          up
Serial0/2/1        unassigned      YES unset   administratively down down
Vlan1            unassigned      YES unset   administratively down down
CentralRouter#
```

Fig 2 – Ip interface brief command output

Department Router

The department router needs ip addressing of the 5 departments [design, project management , finance, HR and IT]. The hostname was only added after the ip addressing between department and central router.

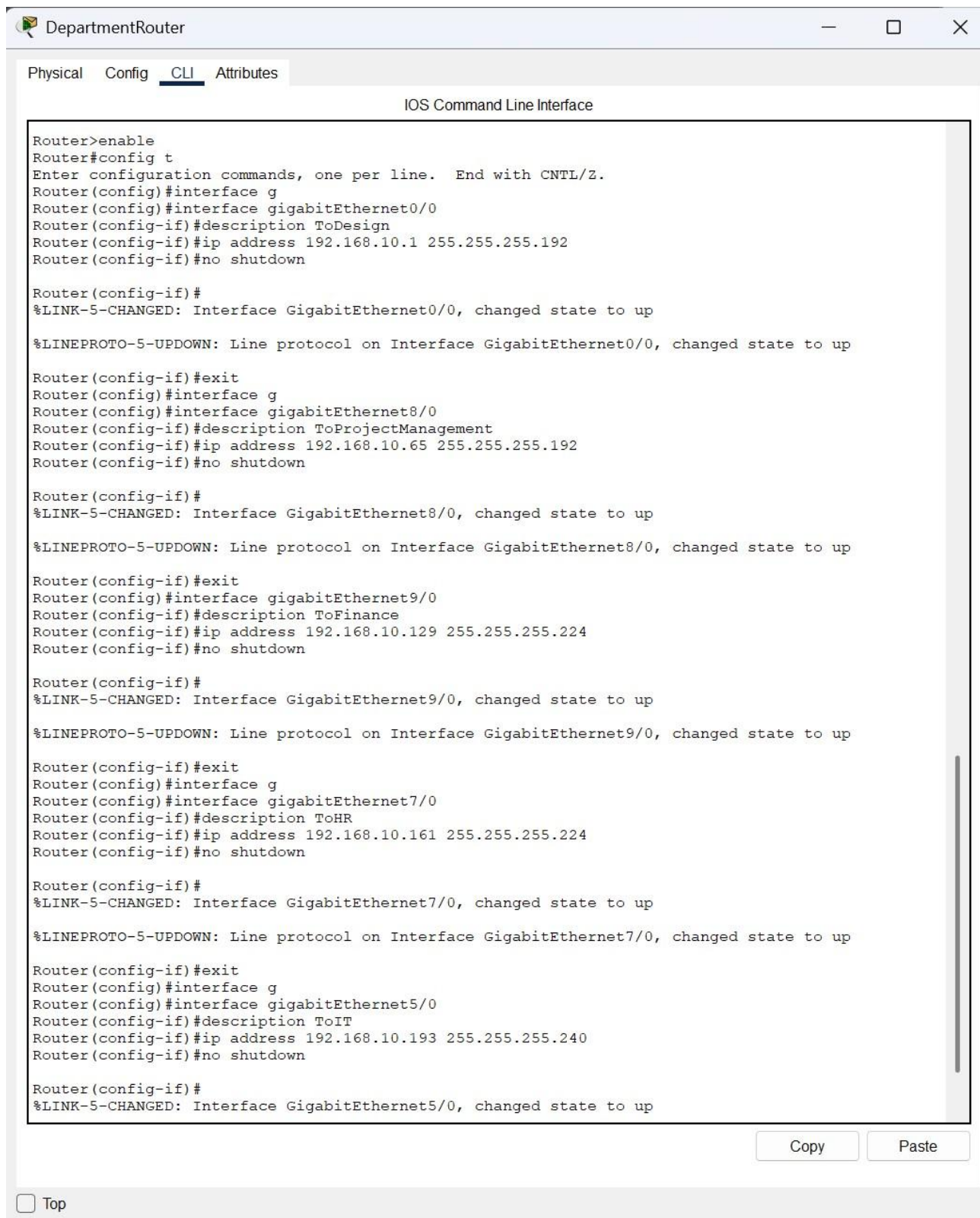


Fig 3 – Ip addressing in department router of the Design, PM, Finance, HR and IT

```

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname DepartmentRouter
DepartmentRouter(config)#interface s
DepartmentRouter(config)#interface serial6/0
DepartmentRouter(config-if)#description ToCentral
DepartmentRouter(config-if)#ip address 192.168.10.241 255.255.255.252
DepartmentRouter(config-if)#no shutdown

DepartmentRouter(config-if)#
%LINK-5-CHANGED: Interface Serial6/0, changed state to up

DepartmentRouter(config-if)#exit
DepartmentRouter(config)#

```

Fig 4 – Ip addressing in central router between Depart and Central Router

Outcome

```

DepartmentRouter#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.10.1	YES	manual	up	up
Modem1/0	unassigned	YES	unset	administratively down	down
Ethernet2/0	unassigned	YES	unset	administratively down	down
FastEthernet3/0	unassigned	YES	unset	administratively down	down
FastEthernet4/0	unassigned	YES	unset	administratively down	down
GigabitEthernet5/0	192.168.10.193	YES	manual	up	up
Serial6/0	192.168.10.241	YES	manual	up	up
GigabitEthernet7/0	192.168.10.161	YES	manual	up	up
GigabitEthernet8/0	192.168.10.65	YES	manual	up	up
GigabitEthernet9/0	192.168.10.129	YES	manual	up	up

Fig 5 – show ip interface brief in Department Router

Server Router

The server router connects to the central router and to the servers [Web, DNS, DHCP, AAA, Email].

```

ServerRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
ServerRouter(config)#interface gigabitEthernet0/0/0
ServerRouter(config-if)#description ToServer
ServerRouter(config-if)#ip address 192.168.10.209 255.255.255.240
ServerRouter(config-if)#no shutdown

ServerRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

ServerRouter(config-if)#exit
ServerRouter(config)#

```

Fig 6 – IP addressing to the servers

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

ServerRouter(config-if)#exit
ServerRouter(config)#interface s
ServerRouter(config)#interface serial0/1/0
ServerRouter(config-if)#description ToCentral
ServerRouter(config-if)#ip address 192.168.10.246 255.255.255.252
ServerRouter(config-if)#no shutdown

ServerRouter(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

ServerRouter(config-if)#exit
ServerRouter(config)#
```

Fig 7 – IP addressing to Server Router

Outcome

The 'show ip interface brief' shows all the ip addresses in the ServerRouter.

```
ServerRouter#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0    192.168.10.209 YES manual up          up
GigabitEthernet0/0/1    unassigned      YES unset  administratively down down
GigabitEthernet0/0/2    unassigned      YES unset  administratively down down
Serial0/1/0              192.168.10.246 YES manual up          up
Serial0/1/1              unassigned      YES unset  administratively down down
GigabitEthernet0/2/0    unassigned      YES unset  up          down
GigabitEthernet0/2/1    unassigned      YES unset  up          down
GigabitEthernet0/2/2    unassigned      YES unset  up          down
GigabitEthernet0/2/3    unassigned      YES unset  up          down
Vlan1                    unassigned      YES unset  administratively down down
ServerRouter#
```

Fig 8 – show IP interface brief in Server Router

OSPF

OSPF protocol is implemented on the DepartmentRouter, CentralRouter and ServerRouter where the network address, wild mask and area code are added. The wild mask is calculated by subtracting [255.255.255.255 – appropriate network address mask].

Subnet mask	Calculation	Wild mask
255.255.255.192	255.255.255.255- 255.255.255.192 =	0.0.0.63
255.255.255.224	255.255.255.255- 255.255.255.224	0.0.0.31
255.255.255.240	255.255.255.255- 255.255.255.240	0.0.0.15
255.255.255.252	255.255.255.255- 255.255.255.252	0.0.0.3

Central Router

```
CentralRouter>enable
CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#router ospf 1
CentralRouter(config-router)#network 192.168.10.240 0.0.0.3 area 0
CentralRouter(config-router)#network 192.168.10.244 0.0.0.3 area 0
CentralRouter(config-router)#exit
CentralRouter(config)#
00:58:44: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.241 on Serial0/1/0 from LOADING to FULL, Loading Done
01:00:02: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.246 on Serial0/2/0 from LOADING to FULL, Loading Done
```

Fig 9 – OSPF setup on central router for the network between department router and server router

```

CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#router ospf 1
CentralRouter(config-router)#network 192.168.10.224 0.0.0.15 area 0
CentralRouter(config-router)#exit
CentralRouter(config)#

```

Fig 10 – adding OSPF for the conference room

Outcome

```

CentralRouter#show running-config | section ospf
router ospf 1
 log-adjacency-changes
 network 192.168.10.240 0.0.0.3 area 0
 network 192.168.10.244 0.0.0.3 area 0
 network 192.168.10.224 0.0.0.15 area 0
CentralRouter#

```

The central router connects the department, server and the conference room. The ospf is important since it allows the router to learn and update routes.

Department Router

```

DepartmentRouter>enable
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#router ospf 1
DepartmentRouter(config-router)#network 192.168.10.0 0.0.0.63 area 0
DepartmentRouter(config-router)#network 192.168.10.64 0.0.0.63 area 0
DepartmentRouter(config-router)#network 192.168.10.128 0.0.0.31 area 0
DepartmentRouter(config-router)#network 192.168.10.160 0.0.0.31 area 0
DepartmentRouter(config-router)#network 192.168.10.192 0.0.0.15 area 0
DepartmentRouter(config-router)#network 192.168.10.240 0.0.0.3 area 0
DepartmentRouter(config-router)#
01:00:04: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.245 on Serial6/0 from LOADING to FULL,
Loading Done
DepartmentRouter(config-router)#exit
DepartmentRouter(config)#

```

Fig 11 – OSPF setup on the department router

Outcome

```

DepartmentRouter#show running-config | section ospf
router ospf 1
 log-adjacency-changes
 network 192.168.10.0 0.0.0.63 area 0
 network 192.168.10.64 0.0.0.63 area 0
 network 192.168.10.128 0.0.0.31 area 0
 network 192.168.10.160 0.0.0.31 area 0
 network 192.168.10.192 0.0.0.15 area 0
 network 192.168.10.240 0.0.0.3 area 0
DepartmentRouter#

```

Fig 12 – OSPF section in the Department Router

Server Router

```
ServerRouter>enable
ServerRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
ServerRouter(config)#router ospf 1
ServerRouter(config-router)#network 192.168.10.208 0.0.0.15 area 0
ServerRouter(config-router)#network 192.168.10.244 0.0.0.3 area 0
ServerRouter(config-router)#
01:26:46: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.245 on Serial0/1/0 from LOADING to FULL,
Loading Done

ServerRouter(config-router)#exit
ServerRouter(config)#
```

Fig 13 – OSPF setup for the server router

Outcome

```
ServerRouter#show running-config | section ospf
router ospf 1
 log-adjacency-changes
 network 192.168.10.208 0.0.0.15 area 0
 network 192.168.10.244 0.0.0.3 area 0
ServerRouter#
```

Fig 14 – OSPF in Server Router

Justification

There are different routing protocols as RIP (Routing Information Protocol) , RIPv2 (version2) and OSPF. RIP and RIP v2 has a limit on the number of hops that are allowed, which are 15. This makes it difficult to implement in larger networks, especially since DesX Zolutions is a growing company. OSPF (Open Shortest Path First), is a link state protocol that is used for large networks. Since it is a widely used protocol and does not have a limit on the number of hops this is more adaptable to use in this network design. For the network to communicate, OSPF needs to be set up in the CentralRouter, DepartmentRouter and the ServerRouter. On the department Router, the network address of the department and address to the CentralRouter along with their respective wildmask is added. On the CentralRouter, the address to the DepartmentRouter and CentralRouter are added to facilitate communication to the respective routers. The address of the conference room is also added since other departments must be able to access it. Lastly, On the ServerRouter the address of the server and address of the CentralRouter are added.

DHCP

DHCP server automatically assigns the ip address for the devices placed under. The DHCP Server is placed in the server router where a static address of 192.168.10.213 is given under the network address of 192.168.10.208 /28.

The screenshot shows the DHCP configuration window with the following settings:

- IP Configuration:** Static IP Address: 192.168.10.213, Subnet Mask: 255.255.255.240, Default Gateway: 192.168.10.209, DNS Server: 0.0.0.0
- IPv6 Configuration:** Static Link Local Address: FE80::2D0:58FF:FE94:5559
- 802.1X:** Use 802.1X Security: ☐ Authentication: MD5, Username: , Password:

Fig 15 – Assign Static address to Server

The screenshot shows the DHCP configuration window with the following settings:

- SERVICES:** DHCP is selected.
- Interface:** FastEthernet0, Service: On
- Pool Name:** ConferenceRoom
- Default Gateway:** 192.168.10.225
- DNS Server:** 192.168.10.211
- Start IP Address:** 192.168.10.226
- Subnet Mask:** 255.255.255.240
- Maximum Number of Users:** 6
- TFTP Server:** 0.0.0.0
- WLC Address:** 0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	V Ad
ConferenceRoom	192.168.10.225	192.168.10.211	192.168.10.226	255.255.255.240	6	0.0.0.0	0.0.0.0
ITDepart	192.168.10.225	192.168.10.211	192.168.10.226	255.255.255.240	10	0.0.0.0	0.0.0.0
HRDepart	192.168.10.225	192.168.10.211	192.168.10.226	255.255.255.224	25	0.0.0.0	0.0.0.0
DesignDepart	192.168.10.1	192.168.10.225	192.168.10.6	255.255.255.192	55	0.0.0.0	0.0.0.0
FinanceDepart	192.168.10.225	192.168.10.211	192.168.10.226	255.255.255.224	30	0.0.0.0	0.0.0.0
PM-Depart	192.168.10.225	192.168.10.211	192.168.10.226	255.255.255.192	40	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.10.226	255.255.255.240	512	0.0.0.0	0.0.0.0

Fig 16 – Assign DHCP service for all the departments and conference room according to requirements

A DHCP pool is created where you add the address from the IP addressing table based on the department requirements.

```

DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#interface g
DepartmentRouter(config)#interface gigabitEthernet0/0
DepartmentRouter(config-if)#ip helper-address 192.168.10.213
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#interface gigabitEthernet8/0
DepartmentRouter(config-if)#ip helper-address 192.168.10.213
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#interface gigabitEthernet9/0
DepartmentRouter(config-if)#ip helper-address 192.168.10.213
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#interface gigabitEthernet7/0
DepartmentRouter(config-if)#ip helper-address 192.168.10.213
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#interface gigabitEthernet5/0
DepartmentRouter(config-if)#ip helper-address 192.168.10.213
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#

```

Fig 17 – helper address to interfaces

```

CentralRouter>enable
CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#interface g
CentralRouter(config)#interface gigabitEthernet0/0/0
CentralRouter(config-if)#ip helper-address 192.168.10.213
CentralRouter(config-if)#exit
CentralRouter(config)#

```

Fig 18 – helper address to interfaces

In the department and central router, 'ip helper-address' is added to their respective interfaces. This helps to forward the broadcast packets since this traffic does not go to the subnets by default.

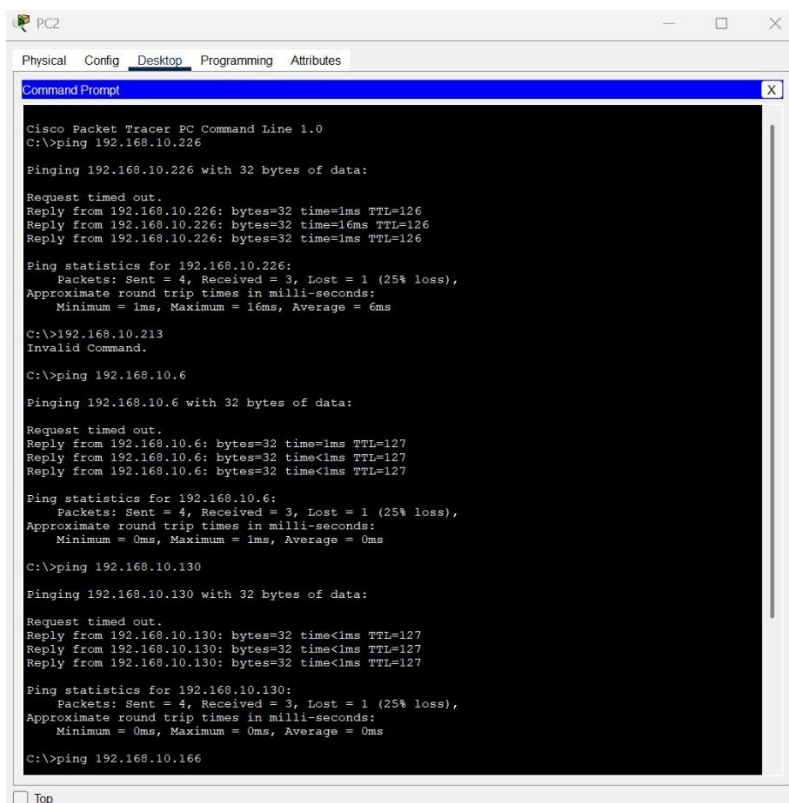


Fig 19 – A PC in pinging different departments

Justification

All the departments are configured using DHCP by adding their respective default-gateway, DNS-Server, Starting IP address, subnet mask along with the maximum number of hosts allowed. This is completed with the routing table and requirements based on each department. As the DHCP server is in another subnet, a helper address is added to the respective interface. This is done in the DepartmentRouter and The CentralRouter. This allows all the departments to automatically receive IP configurations.

The conference room is also configured using DHCP since the server is central and a conference room is like any other department with laptops, wireless devices where every department can access it.

Web/HTTP Server

The Web server is a server placed in the ServerRouter which displays webpages.

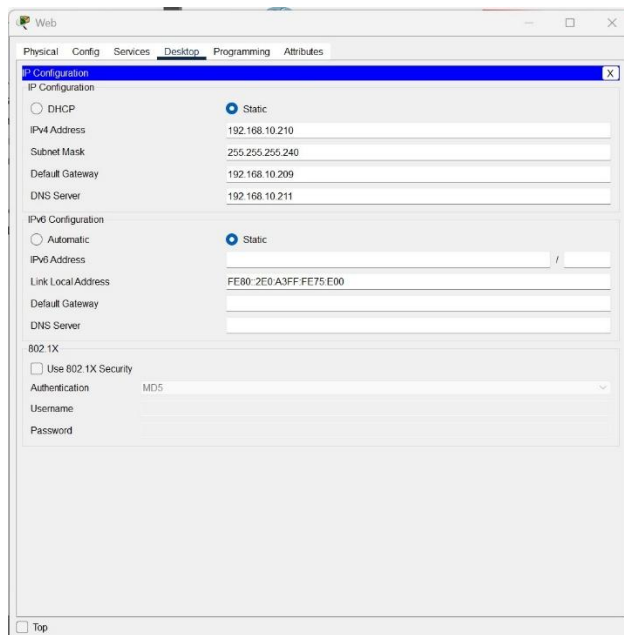


Fig 20 – Assigning Static IP address to web server

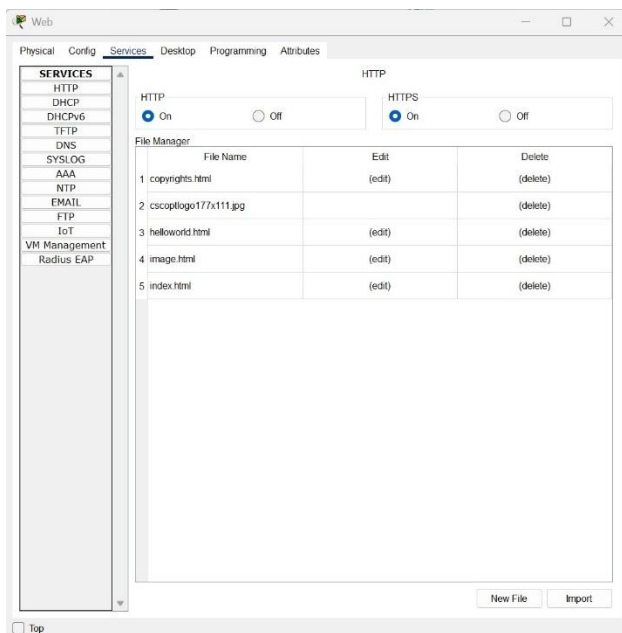


Fig 21 – Switching on the HTTP and HTTPS protocol

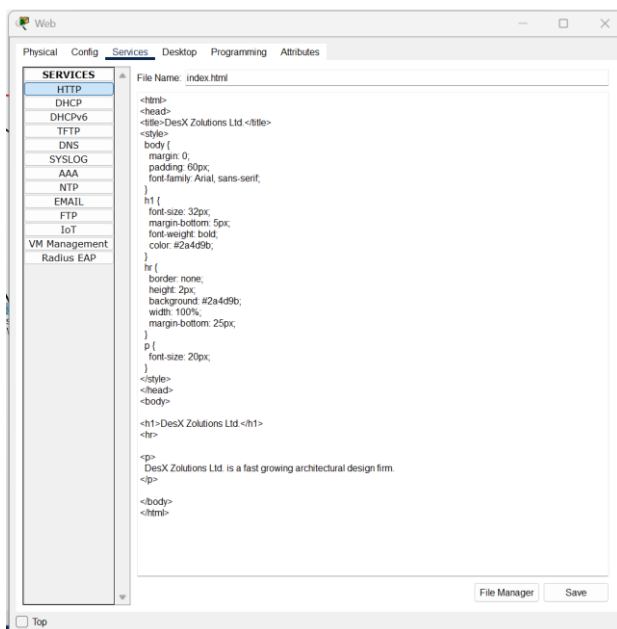


Fig 22 – Adding personalised HTML code for the web page

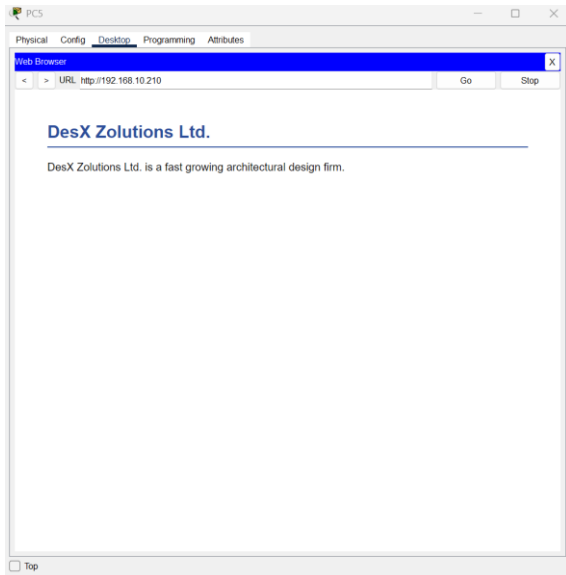


Fig 23 – Web page displayed in http protocol

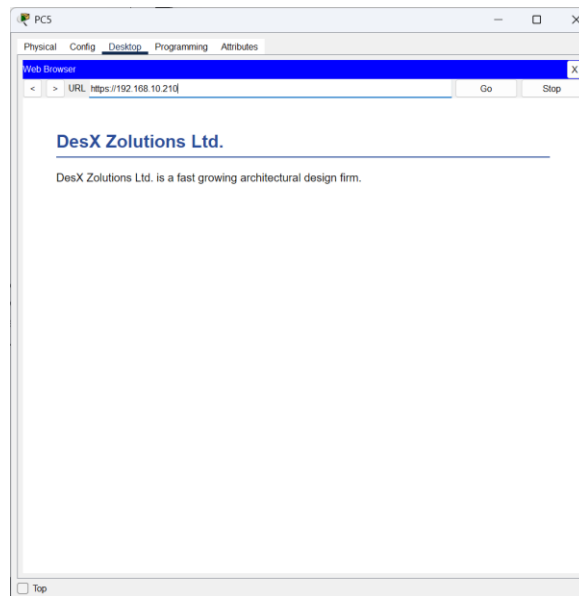


Fig 24 – Web page displayed in https protocol

Justification

The web server helps to display web pages on the device's browser. Initially a static IP address is added along with an DNS address on the web server. For the web browser to display, it is necessary to switch on the http and https services. HTTP is a standard protocol used for loading webpages and HTTPS is the secure version of HTTP. In the http services, in the index.html, html code is added to display a personalised webpage. When a user types in the IP address with http or https, the web browser will be displayed.

DNS Server

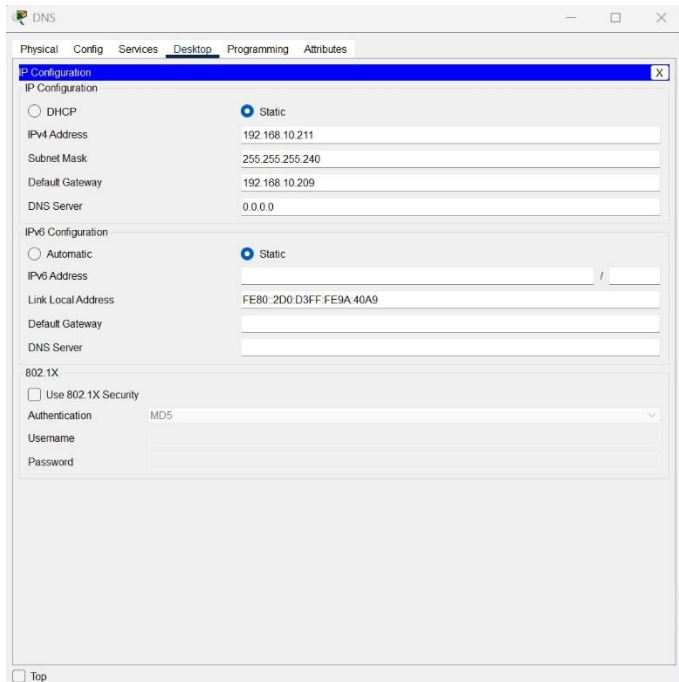


Fig 25 – Assigning Static IP address to dns server

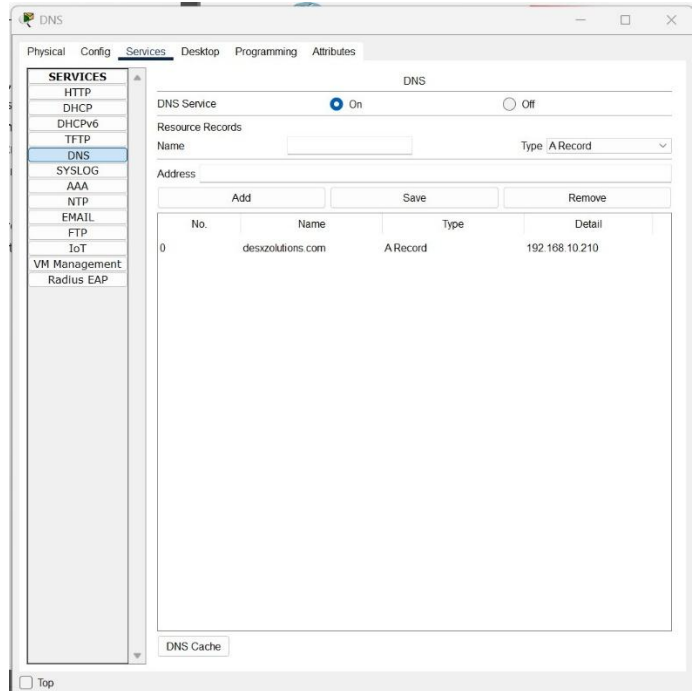


Fig 26 – Switching on the DNS Server

In the DNS service, a name is set 'desxzolutions.com' so it is easy for users to look it up. Then the IP address of the web server is added (192.168.10.210).

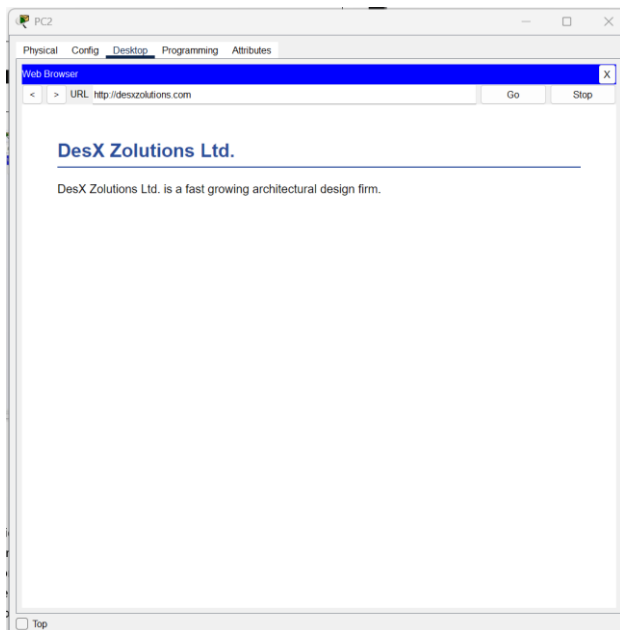


Fig 27 – desxzolutions.com in https protocol

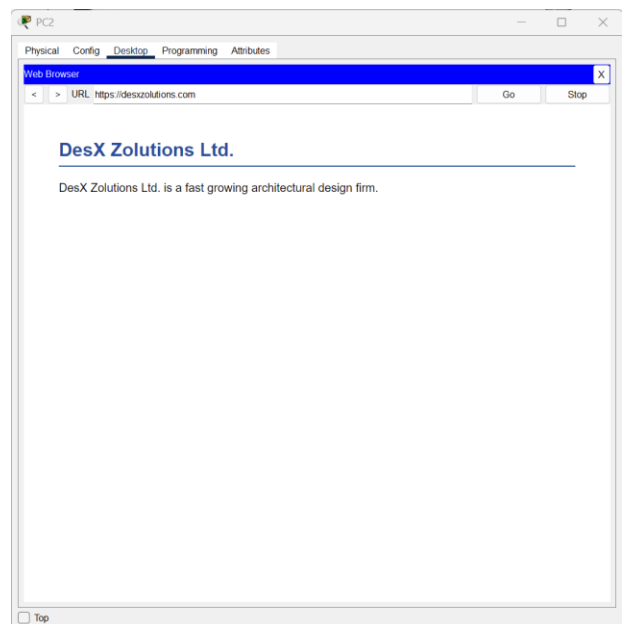


Fig 28 – desxzolutions.com in https protocol

Justification

The DNS server translates ip addresses into domain names. It is quite inconvenient to remember ip address to display web browsers, so a domain name is set (desxzolutions.com). This address is also added in the DHCP services where the pc's will automatically be assigned ip addresses along with the dns server ip address. When a user types in 'desxzolutions.com' it displays the webpage. It displays the webpage in http and https protocol.

Email Server

The email server is used to add the services of email so PC from one department is able to send out an email to another department. The email server is given a static address of (192.168.10.212).

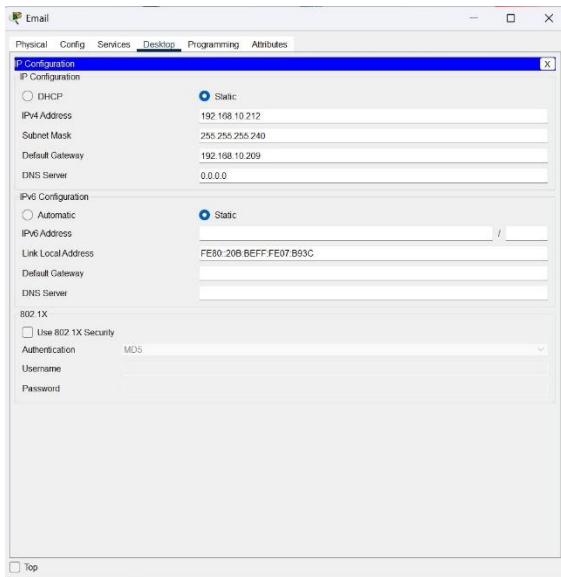


Fig 29 – static address of email server

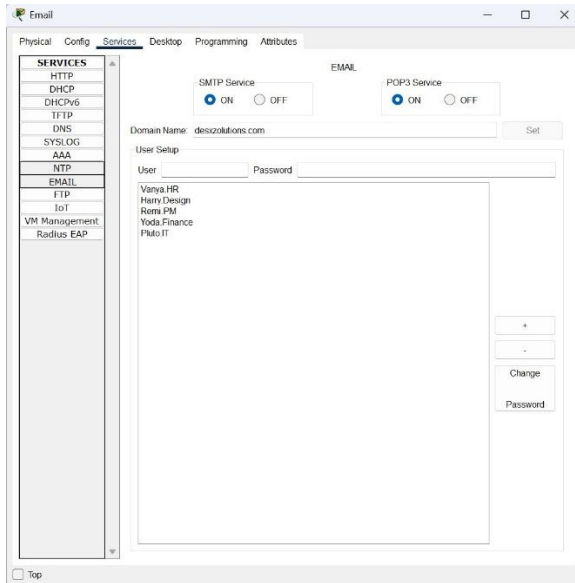


Fig 30 – services of email is added

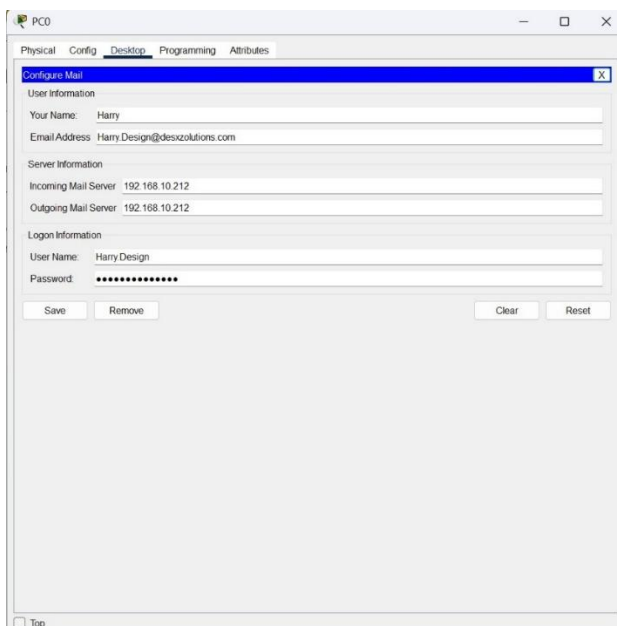


Fig 31 – In PC's, the email account is configured

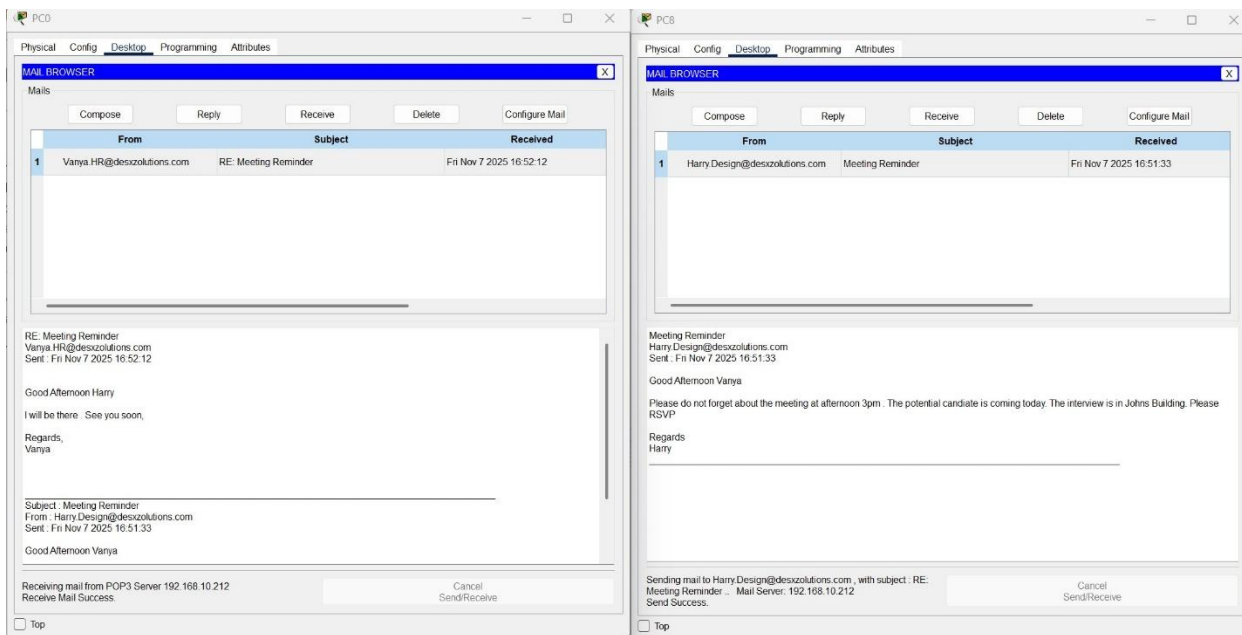


Fig 32 – Email service between a PC in IT and HR

Justification

To use the email services, the SMTP (Simple mail transfer protocol) and POP3 (Post office protocol version3) are switched on. The SMTP is used for sending out mails and POP3 is for receiving mails. Both services are needed in this server. For the users to use an email, their accounts need to be setup. A username, password and a domain name which is 'desxzolutions.com'. All of the users have different password of a combination of uppercase and lowercase letters, numbers and special characters. In the respective PC's, it is necessary to configure their email account. All of the details must be same, in the email of the pc and in the server for the service to work. Once the accounts are configured, emails can be sent.

Wireless Access Point

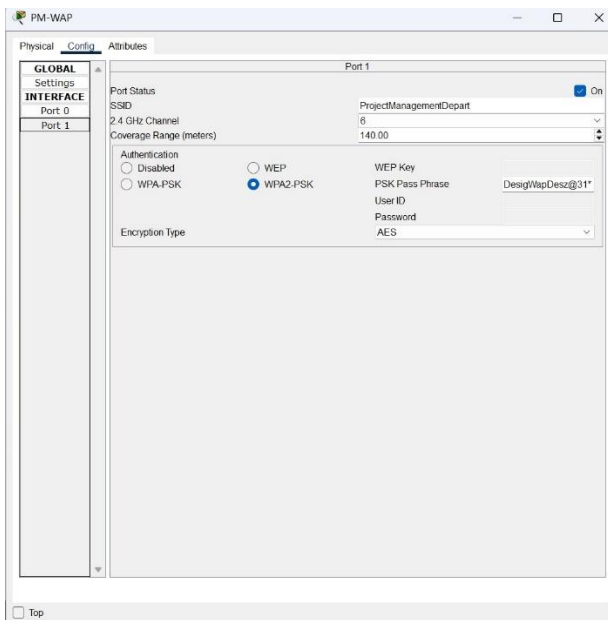


Fig 33 – WAP Port 1 configuration

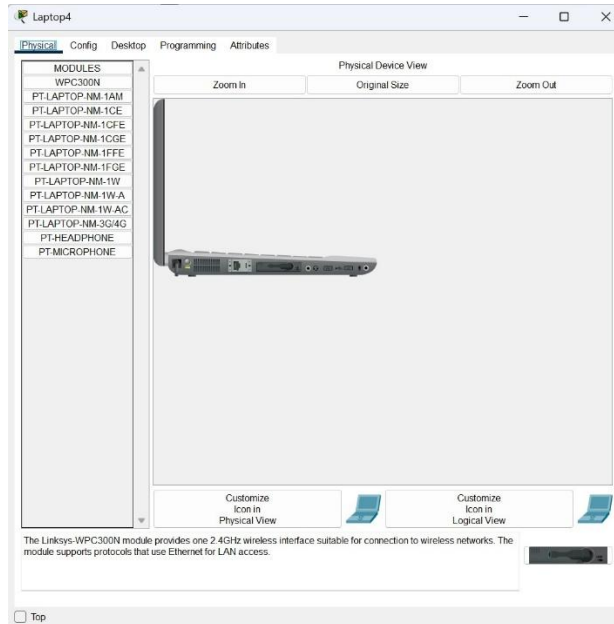


Fig 34 – The WPC300N module dragged in the laptop

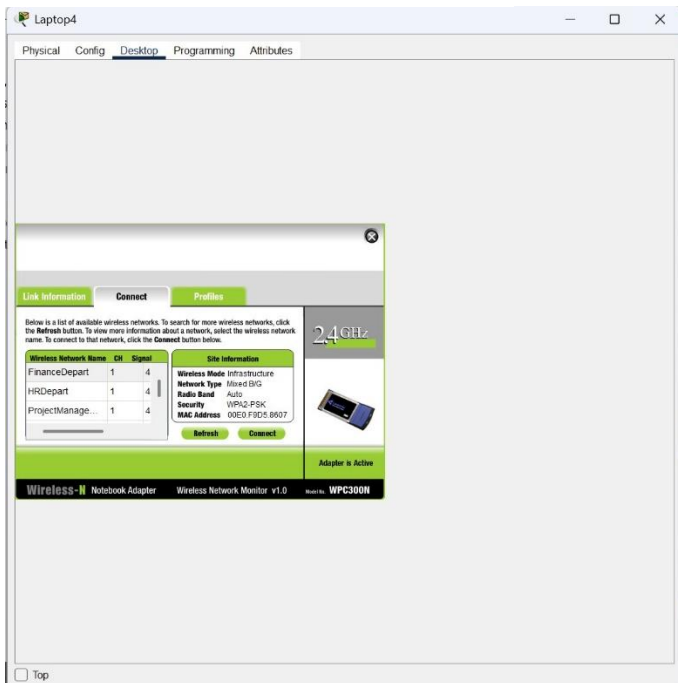


Fig 35 – Different Wireless Network Availability

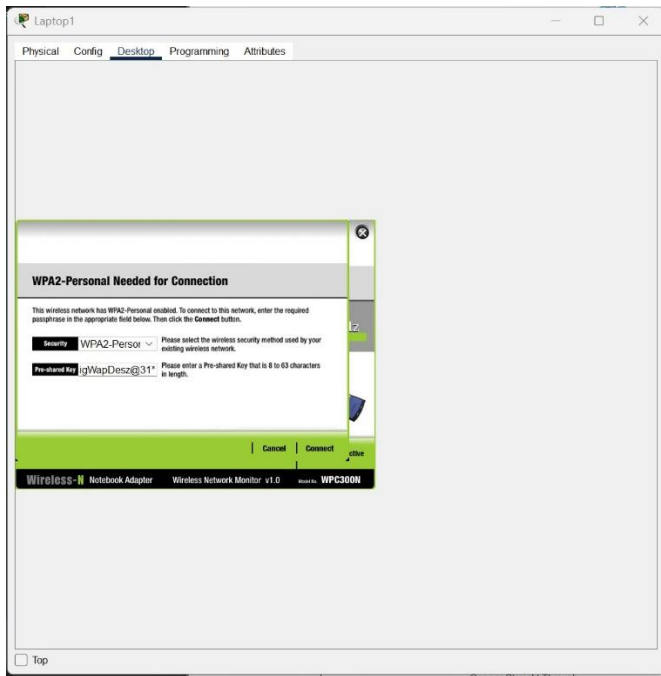


Fig 36 – Password needed for specific WAP

Justification

Every department has its own wireless access point (WAP). This helps users in their respective department to connect to the network using wireless devices like laptops or tablets. For the laptop to connect, it is important to drag the WPC-300N module. This lets the device to connect to a WAP.

In the WAP, in Config, under port 1, a SSID is given which is a name for the WAP which is displayed with other networks when a device is connecting. Among the authentication, WPA2-PSK is selected. This is a stronger security module in comparison to older standards like WEP or WPA. A unique password with a combination of upper and lowercase letter, special symbols and numbers are created. It is important to remember that no same WAP should share the same password.

On the laptop, under the desktop tab, the available wireless networks for each department are displayed. By selecting the appropriate network and entering the correct password, the user can successfully establish a connection, otherwise the connection attempt is rejected.

External Internet Connectivity

The network requires external connectivity which is simulated by using a public IP range of 8.10.1.0/24. The ISP Provider is connected with the central router using a serial cable.

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP-Provider
ISP-Provider(config)#interface s
ISP-Provider(config)#interface serial0/1/0
ISP-Provider(config-if)#description ToCentralRouter
ISP-Provider(config-if)#ip address 8.10.1.1 255.255.255.0
ISP-Provider(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
ISP-Provider(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
|
```

Fig 37 – Adding external network IP address

```
CentralRouter>enable
CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#interface s
CentralRouter(config)#interface serial0/1/1
CentralRouter(config-if)#description ToISP-Provider
CentralRouter(config-if)#ip address 8.10.1.2 255.255.255.0
CentralRouter(config-if)#no shutdown

CentralRouter(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to up

CentralRouter(config-if)#exit
CentralRouter(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
```

Fig 38 – Adding external address on the Central Router

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

ISP-Provider(config-if)#exit
ISP-Provider(config)#ip route 192.168.10.0 255.255.255.0 8.10.1.2
ISP-Provider(config)#|
```

Fig 39 – Adding on ISP

```
CentralRouter(config-if)#exit
CentralRouter(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

CentralRouter(config)#ip route 0.0.0.0 0.0.0.0 8.10.1.1
CentralRouter(config)#
```

Fig 40 – Adding default route in CentralRouter

```
ServerRouter>enable
ServerRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
ServerRouter(config)#ip route 0.0.0.0 0.0.0.0 192.168.10.245
ServerRouter(config)#
```

Fig 41 – Adding default route in ServerRouter

```
DepartmentRouter>enable
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#ip route 0.0.0.0 0.0.0.0 192.168.10.242
DepartmentRouter(config)#
```

Fig 42 – Adding default route in DepartmentRouter

Testing

```
ISP-Provider>enable
ISP-Provider#ping 192.168.10.65

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.65, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 17/22/36 ms

ISP-Provider#ping 192.168.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 21/21/23 ms

ISP-Provider#ping 192.168.10.210

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.210, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 11/17/22 ms

ISP-Provider#ping 8.10.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.10.1.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 6/14/36 ms

ISP-Provider#
```

Fig 43 – ping internal departments in the ISP Provider

Justification

The ISP Provider has a network address of 8.10.1.0 /24. On the ISP Provider interface an IP address of 8.10.1.1/24 is configured, and it is connected to the CentralRouter with an address of 8.10.1.2/24. On the central router, it ensures that all external traffic is forwarded to the ISP Provider.

On the DepartmentRouter and ServerRouter any unfamiliar traffic is forwarded to the CentralRouter. In the ISP 'ip route' command is passed so it knows where to reach the internal network.

Testing

Simulation of message sent from PC in Project Management to a PC in Conference Room

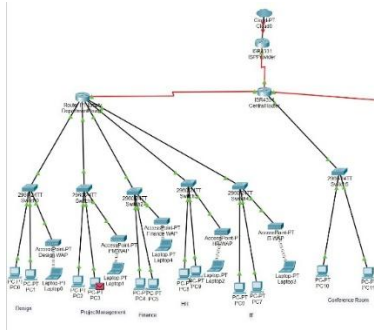


Fig 44- message from pc in PM

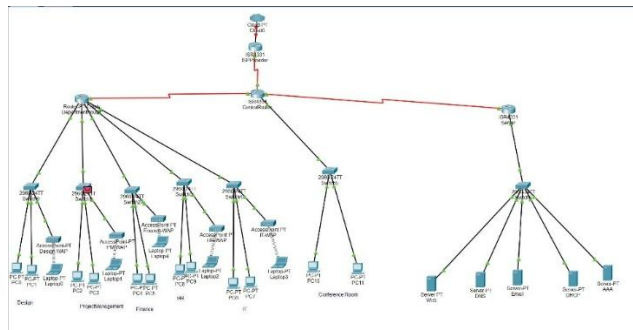


Fig 45- message goes to the switch

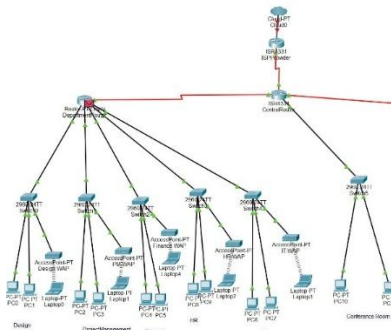


Fig 46- message reaches the departmentrouter

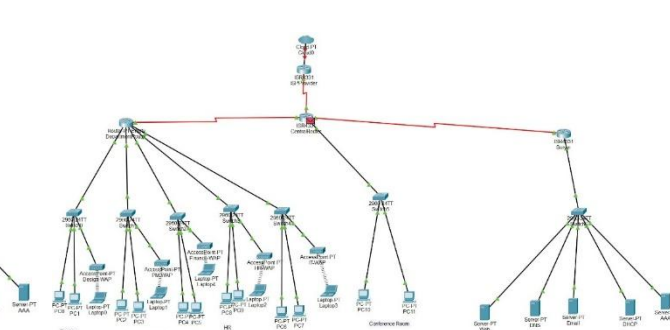


Fig 47- message reaches the central router

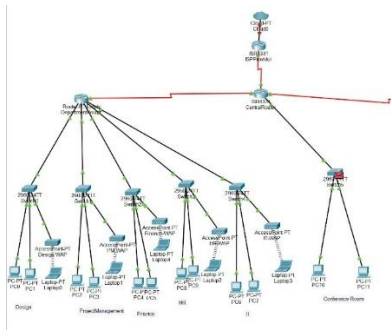


Fig 48- message reaches the switch

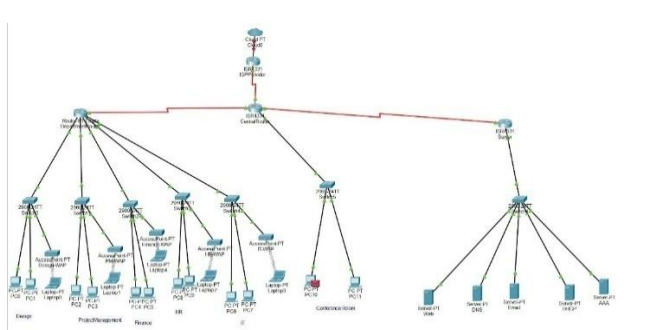


Fig 49- message reaches the pc in conference room

Security Configuration on Routers

```
CentralRouter>enable
CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#no ip domain-lookup
CentralRouter(config)#enable secret Desx@34z
CentralRouter(config)#line console 0
CentralRouter(config-line)#password Cisco34#
CentralRouter(config-line)#login
CentralRouter(config-line)#exit
CentralRouter(config)#line vty 0 4
CentralRouter(config-line)#password Remote@48!
CentralRouter(config-line)#login
CentralRouter(config-line)#exit
CentralRouter(config)#service password-encryption
CentralRouter(config)#banner motd "Unauthorised access to DesX Zolutions Ltd network is prohibited!"
CentralRouter(config)#|
```

```
CentralRouter#show running-config
Building configuration...

Current configuration : 1538 bytes
!
version 16.6.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname CentralRouter
!
!
!
enable secret 5 $1$mERr$8sigvtpy2Bppq1lx17ay.
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
no ip domain-lookup
!
!
!
banner motd ^CUnauthorised access to DesX Zolutions Ltd network is prohibited!^C
!
!
!
line con 0
password 7 0802455D0A16564351
login
!
line aux 0
!
line vty 0 4
password 7 08134943060D003746534D
login
!
!
!
end

CentralRouter#
```

Fig 1 – show running-config CentralRouter

Outcome

```
Building configuration...
Unauthorised access to DesX Zolutions Ltd network is prohibited!

User Access Verification

Password:

CentralRouter>enable
Password:
CentralRouter#config t
Enter configuration commands, one per line.  End with CNTL/Z.
CentralRouter(config)#
```

After the configuration, it now asks for password in the CentralRouter

ISP Provider Router

```
ISP-Provider>enable
ISP-Provider#config t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP-Provider(config)#no ip domain-lookup
ISP-Provider(config)#enable secret Desx@34z
ISP-Provider(config)#line console 0
ISP-Provider(config-line)#password Cisco34#
ISP-Provider(config-line)#login
ISP-Provider(config-line)#exit
ISP-Provider(config)#line vty 0 4
ISP-Provider(config-line)#password Remote@48!
ISP-Provider(config-line)#login
ISP-Provider(config-line)#exit
ISP-Provider(config)#service password-encryption
ISP-Provider(config)#banner motd "Unauthorised access to DesX Zolutins Ltd network is
prohibited!"
ISP-Provider(config)#
```

Fig 2 – basic config on ISP-Provider

```
ISP-Provider#show running-config
Building configuration...

Current configuration : 1301 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname ISP-Provider
!
!
!
enable secret 5 $1$mERr$8sigvtpy2Bppq11x17ay.
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
no ip domain-lookup
!
```

```

!
!
!
banner motd ^CUnauthorised access to DesX Zolutins Ltd network is prohibited!^C
!
!
!
line con 0
password 7 0802455D0A16564351
login
!
line aux 0
!
line vty 0 4
password 7 08134943060D003746534D
login
!
!
!
end

ISP-Provider#|

```

Fig 3 – show running-config ISP-Provider

Outcome

```

Press RETURN to get started!

Unauthorised access to DesX Zolutins Ltd network is prohibited!

User Access Verification

Password:

ISP-Provider>enable
Password:
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

ISP-Provider#config t
Enter configuration commands, one per line. End with CNTL/Z.
ISP-Provider(config)#|

```

The ISP-Provider asks for a password to use the router.

Server Router

```

ServerRouter>enable
ServerRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
ServerRouter(config)#no ip domain-lookup
ServerRouter(config)#enable secret Desx@34z
ServerRouter(config)#line console 0
ServerRouter(config-line)#password Cisco34#
ServerRouter(config-line)#login
ServerRouter(config-line)#exit
ServerRouter(config)#line vty 0 4
ServerRouter(config-line)#password Remote@48!
ServerRouter(config-line)#login
ServerRouter(config-line)#exit
ServerRouter(config)#service password-encryption
ServerRouter(config)#banner motd "Unauthorised access to DesX Zolutions LTd network is prohibited!"
ServerRouter(config)#|

```

Fig 3 – basic config on ServerRouter


```

Press RETURN to get started!

Unauthorised access to DesX Zolutions LTd network is prohibited!

User Access Verification

Password:
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

00:00:30: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.245 on Serial0/1/0 from LOADING to FULL, Loading Done

Password:
Password:

ServerRouter>enable
Password:
ServerRouter#config t

```

DepartmentRouter

```

DepartmentRouter>enable
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#no ip domain-lookup
DepartmentRouter(config)#enable secret Desx@34z
DepartmentRouter(config)#line console 0
DepartmentRouter(config-line)#password Cisco34#
DepartmentRouter(config-line)#login
DepartmentRouter(config-line)#exit
DepartmentRouter(config)#line vty 0 4
DepartmentRouter(config-line)#password Remote@48!
DepartmentRouter(config-line)#login
DepartmentRouter(config-line)#exit
DepartmentRouter(config)#service password-encryption
DepartmentRouter(config)#banner motd "Unauthorised access to DesX Zolutions Ltd network is prohibited!"
DepartmentRouter(config)#

```

Fig 3 – config on DepartmentRouter

```

DepartmentRouter#show running-config
Building configuration...

Current configuration : 2037 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DepartmentRouter
!
!
!
enable secret 5 $1$mERr$8sigvtpy2Bppq11x17ay.
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
no ip domain-lookup
!

```

```

:
!
banner motd ^CUnauthorised access to DesX Zolutions Ltd network is prohibited!^C
!
!
!
!
!
line con 0
 password 7 0802455D0A16564351
 login
!
line aux 0
!
line vty 0 4
 password 7 08134943060D003746534D
 login
!
!
!
end

DepartmentRouter#
DepartmentRouter#
DepartmentRouter#

```

Fig 3 – show running-config on DepartmentRouter

Outcome

```

Press RETURN to get started!

Unauthorised access to DesX Zolutions Ltd network is prohibited!

User Access Verification

Password:
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial6/0, changed state to up

DepartmentRouter>enable
Password:
00:00:26: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.245 on Serial6/0 from LOADING to FULL,
Loading Done

DepartmentRouter#config t
Enter configuration commands, one per line.  End with CNTL/Z.
DepartmentRouter(config)#

```

Justification

In the network, every router is configured with an appropriate hostname, encrypted enable password and disable ip domain-lookup. The line console and line VTY is configured on the routers so only authenticated users can access. A login banner “Unauthorised access to DesX Zolutions Ltd network is prohibited!” is displayed. This is to show a warning about unauthorised users trying to access the devices. The command ‘service password-encryption’ stores the passwords encrypted which provides an additional security to the device. The servers use a consistent password as shown below.

Console password	Cisco34#
Enable password	Desx@34z
VTY password	Remote@48!
Localadmin	backupadmin02

AAA

Before configuring AAA, a Localadmin backup account is added to the routers in case the AAA locks the device out.

Central Router

```
CentralRouter>enable
Password:
CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#username Localadmin privilege 15 secret backupadmin02
CentralRouter(config)#end
CentralRouter#
%SYS-5-CONFIG_I: Configured from console by console

CentralRouter#wr
Building configuration...
[OK]
CentralRouter#
```

```
CentralRouter#wr
Building configuration...
[OK]
CentralRouter#show running-config
Building configuration...

Current configuration : 1611 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname CentralRouter
!
!
!
enable secret 5 $1$mERr$8sigvtppy2Bppq1lx17ay.
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username Localadmin privilege 15 secret 5 $1$mERr$3xztuoXhh81ym9Y8G0DV31
!
!
!
!
!
```

Fig 4 – Localadmin configured and seen by 'show running-config' in CentralServer

Department Router

```
-----
Password:
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#username Localadmin privilege 15 secret backupadmin02
DepartmentRouter(config)#end
DepartmentRouter#
%SYS-5-CONFIG_I: Configured from console by console

DepartmentRouter#wr
Building configuration...
[OK]
DepartmentRouter#
```

```

DepartmentRouter#show running-config
Building configuration...

Current configuration : 2102 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DepartmentRouter
!
!
!
enable secret 5 $1$mERr$8sigvtpy2Bppq11x17ay.
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username Localadmin privilege 15 secret 5 $1$mERr$3xztuoXhh81ym9Y8G0DV31
!
!
!
!

```

Fig 5 – Localadmin configured and seen by 'show running-config' in DepartmentRouter

Server Router

```

User Access Verification

Password:

ServerRouter>enable
Password:
ServerRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
ServerRouter(config)#username Localadmin privilege 15 secret backupadmin02
ServerRouter(config)#end
ServerRouter#
%SYS-5-CONFIG_I: Configured from console by console

ServerRouter#wr
Building configuration...
[OK]
ServerRouter#

```

```

[OK]
ServerRouter#show running-config
Building configuration...

Current configuration : 1532 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname ServerRouter
!
!
!
enable secret 5 $1$mERr$8sigvtppy2Bppq11x17ay.
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username Localadmin privilege 15 secret 5 $1$mERr$3xztuoXhh81ym9Y8G0DV31
!
!
!

```

Fig 6 – Localadmin configured and seen by 'show running-config' in ServerRouter

AAA server

AAA Configuration window showing IP Configuration, IPv6 Configuration, and 802.1X settings.

Fig 7 – AAA ip addressing

In the AAA server, a static ip address is given to start its configuration.

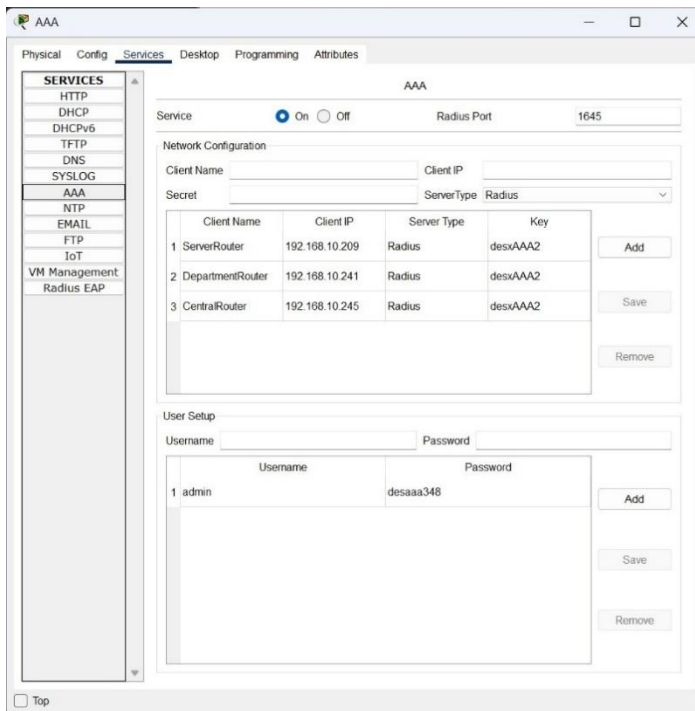


Fig 8 – The ip address of the server , department and central routers are added.

Justification

As a server, the AAA is given a static ip address of 192.168.10.214. In the server, under the services, AAA is configured. The service is switched on and ip address of the Department, Server and Central Routers are added. The ip address of the routers are added along with a secret key 'desxAAA2'. This is added with the server type as RADIUS. An account 'admin' is created under the user setup. When the user now logs into the router, the credentials of AAA (admin) is to be added. The password used for AAA in the routers are the same and the password combination are kept quite simple to test the network. However, in future a good combination of uppercase, lowercase, numbers and special characters must be used by following an appropriate password policy. This ensures enhanced security and reduces the risk of unauthorised access.

Central Router

```
CentralRouter(config)#aaa new-model
CentralRouter(config)#radius server SERVER-AAA
CentralRouter(config-radius-server)#address ipv4 192.168.10.214
CentralRouter(config-radius-server)#key desxAAA2
WARNING: Command has been added to the configuration using a type 0 password. However, type 0
passwords will soon be deprecated. Migrate to a supported password type
*Nov 08 10:09:45.063: %AAAA-4-CLI_DEPRECATED: WARNING: Command has been added to the
configuration using a type 0 password. However, type 0 passwords will soon be deprecated.
Migrate to a supported password type
CentralRouter(config-radius-server)#exit
CentralRouter(config)#aaa authentication login default group radius local
CentralRouter(config)#line con 0
CentralRouter(config-line)#login authentication default
CentralRouter(config-line)#exit
CentralRouter(config)#line vty 0 4
CentralRouter(config-line)#login authentication default
CentralRouter(config-line)#exit
CentralRouter(config)#end
CentralRouter#
%SYS-5-CONFIG_I: Configured from console by console

CentralRouter#wr
Building configuration...
[OK]
CentralRouter#
```

Fig 9 – creating aaa model in central router

```
Loading Done
Unauthorised access to DesX Zolutions Ltd network is prohibited!

User Access Verification

Username: admin
Password:
CentralRouter>
```

Fig 10 – The credentials of AAA are used to log in the CentralRouter

Department Router

In the department Router, the AAA new-model is created to configure the Radius AAA.

```
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#aaa new-model
DepartmentRouter(config)#radius server SERVER-AAA
                        ^
% Invalid input detected at '^' marker.

DepartmentRouter(config)#radius-server host 192.168.10.214
DepartmentRouter(config)#radius-server key desxAAA2
DepartmentRouter(config)#aaa authentication login default group radius local
DepartmentRouter(config)#line con 0
DepartmentRouter(config-line)#login authentication default
DepartmentRouter(config-line)#exit
DepartmentRouter(config)#line vty 0 4
DepartmentRouter(config-line)#login authentication default
DepartmentRouter(config-line)#exit
DepartmentRouter(config)#end
DepartmentRouter#
%SYS-5-CONFIG_I: Configured from console by console

DepartmentRouter#wr
Building configuration...
[OK]
DepartmentRouter#
```

Fig 11 – creating AAA model in DepartmentRouter

```
Press RETURN to get started!

Unauthorised access to DesX Zolutions Ltd network is prohibited!

User Access Verification

Username: admin
Password:
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial6/0, changed state to up

DepartmentRouter>
00:00:26: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.245 on Serial6/0 from LOADING to FULL,
Loading Done
enable
Password:
DepartmentRouter#
```

Fig 12 – AAA credentials are used in the DepartmentRouter

Server Router

```
ServerRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
ServerRouter(config)#aaa new-model
```

```
ServerRouter(config)#radius server SERVER-AAA
ServerRouter(config-radius-server)#address ipv4 192.168.10.214
ServerRouter(config-radius-server)#key desxAAA2
WARNING: Command has been added to the configuration using a type 0 password. However, type 0
passwords will soon be deprecated. Migrate to a supported password type
*Nov 08 09:51:12.349: %AAAA-4-CLI_DEPRECATED: WARNING: Command has been added to the
configuration using a type 0 password. However, type 0 passwords will soon be deprecated.
Migrate to a supported password type
ServerRouter(config-radius-server)#exit
ServerRouter(config)#aaa authentication login default group radius local
ServerRouter(config)#line con 0
ServerRouter(config-line)#login authentication default
ServerRouter(config-line)#exit
ServerRouter(config)#line vty 0 4
ServerRouter(config-line)#login authentication default
ServerRouter(config-line)#exit
ServerRouter(config)#end
ServerRouter#
%SYS-5-CONFIG_I: Configured from console by console

ServerRouter#
```

Fig 13 – creating AAA model in server router

```
Press RETURN to get started!

Unauthorised access to DesX Zolutions LTd network is prohibited!

User Access Verification

Username: admin
Password:
ServerRouter>
```

Fig 14 – logging into the serverrouter using AAA credentials

Justification

AAA (Authentication, Authorisation and Accounting) is a service framework responsible for managing and securing the network. This is configured across the three routers of Department, Central and Server using a Radius (AAA) server. Initially the AAA service is switched on where the ip address of the appropriate servers are added. The user account is also set up in the server, where these credentials will be used to log into the routers. The Radius protocol is used with AAA as it combines the authentication and authorisation while supporting accounting separately. AAA ensures no unauthorised users can make changes, which helps to improve the network security and ensure a more secure access to devices. The network is expected to grow and Radius is a flexible protocol that can be used to secure the network.

ACL

Access control list helps to further define who can access which department. For example, in design only project management and design is allowed access, but the rest of the departments are restricted.

Design department

```
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#ip access-list extended ACL-Design
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.63 192.168.10.64 0.0.0.63
^
% Invalid input detected at '^' marker.

DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.63 192.168.10.64 0.0.0.63
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.63 192.168.10.208 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit udp any eq 68 any eq 67
DepartmentRouter(config-ext-nacl)#permit udp any eq 67 any eq 68
DepartmentRouter(config-ext-nacl)#permit icmp 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15
echo-reply
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.63 192.168.10.128 0.0.0.31
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.63 192.168.10.160 0.0.0.31
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.0 0.0.0.63 any
DepartmentRouter(config-ext-nacl)#exit
DepartmentRouter(config)#interface g
DepartmentRouter(config)#interface gigabitEthernet0/0
DepartmentRouter(config-if)#ip access-group ACL-Design in
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#end
DepartmentRouter#
%SYS-5-CONFIG_I: Configured from console by console

DepartmentRouter#wr
Building configuration...
[OK]
DepartmentRouter#show access-list
Extended IP access list ACL-Design
 10 permit ip 192.168.10.0 0.0.0.63 192.168.10.64 0.0.0.63
 20 permit ip 192.168.10.0 0.0.0.63 192.168.10.208 0.0.0.15
 30 permit udp any eq bootpc any eq bootps
 40 permit udp any eq bootps any eq bootpc
 50 permit icmp 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15 echo-reply
 60 deny ip 192.168.10.0 0.0.0.63 192.168.10.128 0.0.0.31
 70 deny ip 192.168.10.0 0.0.0.63 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.0 0.0.0.63 any
DepartmentRouter#
```

Fig 15 – creating extended ACL for design

Project Management Department

The project management department allows access to design and finance and restricts other departments.

```
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#ip access-list extended ACL-Pm
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.64 0.0.0.63 192.168.10.0 0.0.0.63
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.64 0.0.0.63 192.168.10.128 0.0.0.31
DepartmentRouter(config-ext-nacl)#permit udp any eq 68 any eq 67
DepartmentRouter(config-ext-nacl)#permit udp any eq 67 any eq 68
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.64 0.0.0.63 192.168.10.208 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit icmp 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15
echo-reply
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.64 0.0.0.63 192.168.10.160 0.0.0.31
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.64 0.0.0.63 any
DepartmentRouter(config-ext-nacl)#exit
DepartmentRouter(config)#interface g
DepartmentRouter(config)#interface gigabitEthernet8/0
DepartmentRouter(config-if)#ip access-group ACL-Pm in
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#end
DepartmentRouter#
%SYS-5-CONFIG_I: Configured from console by console

DepartmentRouter#wr
Building configuration...
[OK]
DepartmentRouter#show access-lists
Extended IP access list ACL-Design
 10 permit ip 192.168.10.0 0.0.0.63 192.168.10.64 0.0.0.63
 20 permit ip 192.168.10.0 0.0.0.63 192.168.10.208 0.0.0.15
 30 permit udp any eq bootpc any eq bootps
 40 permit udp any eq bootps any eq bootpc
 50 permit icmp 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15 echo-reply
 60 deny ip 192.168.10.0 0.0.0.63 192.168.10.128 0.0.0.31
 70 deny ip 192.168.10.0 0.0.0.63 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.0 0.0.0.63 any
Extended IP access list ACL-Pm
 10 permit ip 192.168.10.64 0.0.0.63 192.168.10.0 0.0.0.63
 20 permit ip 192.168.10.64 0.0.0.63 192.168.10.128 0.0.0.31
 30 permit udp any eq bootpc any eq bootps
 40 permit udp any eq bootps any eq bootpc
 50 permit ip 192.168.10.64 0.0.0.63 192.168.10.208 0.0.0.15
 60 permit icmp 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15 echo-reply
 70 deny ip 192.168.10.64 0.0.0.63 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.64 0.0.0.63 any
```

Fig 16 – creating extended ACL for P.M

Finance Department

```
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#ip access-list extended ACL-Finance
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.128 0.0.0.31 192.168.10.64 0.0.0.63
DepartmentRouter(config-ext-nacl)#permit udp any eq 68 any eq 67
DepartmentRouter(config-ext-nacl)#permit udp any eq 67 any eq 68
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.128 0.0.0.31 192.168.10.208 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit icmp 192.168.10.128 0.0.0.31 192.168.10.192 0.0.0.15
echo-reply
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.128 0.0.0.31 192.168.10.0 0.0.0.63
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.128 0.0.0.31 192.168.10.160 0.0.0.31
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.128 0.0.0.31 192.168.10.192 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.128 0.0.0.31 any
DepartmentRouter(config-ext-nacl)#exit
DepartmentRouter(config)#interface g
DepartmentRouter(config)#interface gigabitEthernet9/0
DepartmentRouter(config-if)#ip access-group ACL-Finance in
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#end
DepartmentRouter#
%SYS-5-CONFIG_I: Configured from console by console

DepartmentRouter#wr
Building configuration...
[OK]
DepartmentRouter#show access-lists
Extended IP access list ACL-Design
 10 permit ip 192.168.10.0 0.0.0.63 192.168.10.64 0.0.0.63
 20 permit ip 192.168.10.0 0.0.0.63 192.168.10.208 0.0.0.15
 30 permit udp any eq bootpc any eq bootps
 40 permit udp any eq bootps any eq bootpc
 50 permit icmp 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15 echo-reply
 60 deny ip 192.168.10.0 0.0.0.63 192.168.10.128 0.0.0.31
 70 deny ip 192.168.10.0 0.0.0.63 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.0 0.0.0.63 any
Extended IP access list ACL-Fm
 10 permit ip 192.168.10.64 0.0.0.63 192.168.10.0 0.0.0.63
 20 permit ip 192.168.10.64 0.0.0.63 192.168.10.128 0.0.0.31
 30 permit udp any eq bootpc any eq bootps
 40 permit udp any eq bootps any eq bootpc
 50 permit ip 192.168.10.64 0.0.0.63 192.168.10.208 0.0.0.15
 60 permit icmp 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15 echo-reply
 70 deny ip 192.168.10.64 0.0.0.63 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.64 0.0.0.63 any
Extended IP access list ACL-Finance
 10 permit ip 192.168.10.128 0.0.0.31 192.168.10.64 0.0.0.63
 20 permit udp any eq bootpc any eq bootps
 30 permit udp any eq bootps any eq bootpc
 40 permit ip 192.168.10.128 0.0.0.31 192.168.10.208 0.0.0.15
 50 permit icmp 192.168.10.128 0.0.0.31 192.168.10.192 0.0.0.15 echo-reply
 60 deny ip 192.168.10.128 0.0.0.31 192.168.10.0 0.0.0.63
 70 deny ip 192.168.10.128 0.0.0.31 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.128 0.0.0.31 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.128 0.0.0.31 any
DepartmentRouter#
```

Fig 17 – creating extended ACL for Finance

HR Department

```
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#ip access-list extended ACL-Hr
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.160 0.0.0.31 192.168.10.208 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit udp any eq 68 any eq 67
DepartmentRouter(config-ext-nacl)#permit udp any eq 67 any eq 68
DepartmentRouter(config-ext-nacl)#permit icmp 192.168.10.160 0.0.0.31 192.168.10.192 0.0.0.15
echo-reply
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.160 0.0.0.31 192.168.10.0 0.0.0.63
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.160 0.0.0.31 192.168.10.64 0.0.0.63
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.160 0.0.0.31 192.168.10.128 0.0.0.31
DepartmentRouter(config-ext-nacl)#deny ip 192.168.10.160 0.0.0.31 192.168.10.192 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.160 0.0.0.31 any
DepartmentRouter(config-ext-nacl)#exit
DepartmentRouter(config)#interface g
DepartmentRouter(config)#interface gigabitEthernet7/0
DepartmentRouter(config-if)#ip access-group ACL-Hr in
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#end
DepartmentRouter#
%SYS-5-CONFIG_I: Configured from console by console

DepartmentRouter#wr
Building configuration...
[OK]
DepartmentRouter#show access-lists
Extended IP access list ACL-Design
 10 permit ip 192.168.10.0 0.0.0.63 192.168.10.64 0.0.0.63
 20 permit ip 192.168.10.0 0.0.0.63 192.168.10.208 0.0.0.15
 30 permit udp any eq bootpc any eq bootps
 40 permit udp any eq bootps any eq bootpc
 50 permit icmp 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15 echo-reply
 60 deny ip 192.168.10.0 0.0.0.63 192.168.10.128 0.0.0.31
 70 deny ip 192.168.10.0 0.0.0.63 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.0 0.0.0.63 any
Extended IP access list ACL-Fm
 10 permit ip 192.168.10.64 0.0.0.63 192.168.10.0 0.0.0.63
 20 permit ip 192.168.10.64 0.0.0.63 192.168.10.128 0.0.0.31
 30 permit udp any eq bootpc any eq bootps
 40 permit udp any eq bootps any eq bootpc
 50 permit ip 192.168.10.64 0.0.0.63 192.168.10.208 0.0.0.15
 60 permit icmp 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15 echo-reply
 70 deny ip 192.168.10.64 0.0.0.63 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.64 0.0.0.63 any
Extended IP access list ACL-Finance
 10 permit ip 192.168.10.128 0.0.0.31 192.168.10.64 0.0.0.63
 20 permit udp any eq bootpc any eq bootps
 30 permit udp any eq bootps any eq bootpc
 40 permit ip 192.168.10.128 0.0.0.31 192.168.10.208 0.0.0.15
 50 permit icmp 192.168.10.128 0.0.0.31 192.168.10.192 0.0.0.15 echo-reply
 60 deny ip 192.168.10.128 0.0.0.31 192.168.10.0 0.0.0.63
 70 deny ip 192.168.10.128 0.0.0.31 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.128 0.0.0.31 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.128 0.0.0.31 any
Extended IP access list ACL-Hr
 10 permit ip 192.168.10.160 0.0.0.31 192.168.10.208 0.0.0.15
 20 permit udp any eq bootpc any eq bootps
 30 permit udp any eq bootps any eq bootpc
 40 permit icmp 192.168.10.160 0.0.0.31 192.168.10.192 0.0.0.15 echo-reply
 50 deny ip 192.168.10.160 0.0.0.31 192.168.10.0 0.0.0.63
 60 deny ip 192.168.10.160 0.0.0.31 192.168.10.64 0.0.0.63
 70 deny ip 192.168.10.160 0.0.0.31 192.168.10.128 0.0.0.31
 80 deny ip 192.168.10.160 0.0.0.31 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.160 0.0.0.31 any
```

Fig 18 – creating extended ACL for HR

IT department

```
DepartmentRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
DepartmentRouter(config)#ip access-list extended ACL-IT
DepartmentRouter(config-ext-nacl)#permit udp any eq 68 any eq 67
DepartmentRouter(config-ext-nacl)#permit udp any eq 67 any eq 68
DepartmentRouter(config-ext-nacl)#permit icmp 192.168.10.192 0.0.0.15 192.168.10.0 0.0.0.63
DepartmentRouter(config-ext-nacl)#permit icmp 192.168.10.192 0.0.0.15 192.168.10.64 0.0.0.63
DepartmentRouter(config-ext-nacl)#permit icmp 192.168.10.192 0.0.0.15 192.168.10.128 0.0.0.31
DepartmentRouter(config-ext-nacl)#permit icmp 192.168.10.192 0.0.0.15 192.168.10.160 0.0.0.31
DepartmentRouter(config-ext-nacl)#permit icmp 192.168.10.192 0.0.0.15 192.168.10.208 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.192 0.0.0.15 192.168.10.0 0.0.0.255
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.192 0.0.0.15 192.168.10.208 0.0.0.15
DepartmentRouter(config-ext-nacl)#permit ip 192.168.10.192 0.0.0.15 any
DepartmentRouter(config-ext-nacl)#exit
DepartmentRouter(config)#interface g
DepartmentRouter(config)#interface gigabitEthernet5/0
DepartmentRouter(config-if)#ip access-group ACL-IT in
DepartmentRouter(config-if)#exit
DepartmentRouter(config)#end
DepartmentRouter#
%SYS-5-CONFIG_I: Configured from console by console

DepartmentRouter#wr
Building configuration...
[OK]
DepartmentRouter#show access-lists
```

Fig 19 – creating extended ACL for IT

Outcome

The entire extended ip access-lists are as below

```
DepartmentRouter#show access-lists
Extended IP access list ACL-Design
 10 permit ip 192.168.10.0 0.0.0.63 192.168.10.64 0.0.0.63
 20 permit ip 192.168.10.0 0.0.0.63 192.168.10.208 0.0.0.15
 30 permit udp any eq bootpc any eq bootps
 40 permit udp any eq bootps any eq bootpc
 50 permit icmp 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15 echo-reply
 60 deny ip 192.168.10.0 0.0.0.63 192.168.10.128 0.0.0.31
 70 deny ip 192.168.10.0 0.0.0.63 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.0 0.0.0.63 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.0 0.0.0.63 any
Extended IP access list ACL-Fm
 10 permit ip 192.168.10.64 0.0.0.63 192.168.10.0 0.0.0.63
 20 permit ip 192.168.10.64 0.0.0.63 192.168.10.128 0.0.0.31
 30 permit udp any eq bootpc any eq bootps
 40 permit udp any eq bootps any eq bootpc
 50 permit ip 192.168.10.64 0.0.0.63 192.168.10.208 0.0.0.15
 60 permit icmp 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15 echo-reply
 70 deny ip 192.168.10.64 0.0.0.63 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.64 0.0.0.63 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.64 0.0.0.63 any
Extended IP access list ACL-Finance
 10 permit ip 192.168.10.128 0.0.0.31 192.168.10.64 0.0.0.63
 20 permit udp any eq bootpc any eq bootps
 30 permit udp any eq bootps any eq bootpc
 40 permit ip 192.168.10.128 0.0.0.31 192.168.10.208 0.0.0.15
 50 permit icmp 192.168.10.128 0.0.0.31 192.168.10.192 0.0.0.15 echo-reply
 60 deny ip 192.168.10.128 0.0.0.31 192.168.10.0 0.0.0.63
 70 deny ip 192.168.10.128 0.0.0.31 192.168.10.160 0.0.0.31
 80 deny ip 192.168.10.128 0.0.0.31 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.128 0.0.0.31 any
Extended IP access list ACL-Hr
 10 permit ip 192.168.10.160 0.0.0.31 192.168.10.208 0.0.0.15
 20 permit udp any eq bootpc any eq bootps
 30 permit udp any eq bootps any eq bootpc
 40 permit icmp 192.168.10.160 0.0.0.31 192.168.10.192 0.0.0.15 echo-reply
 50 deny ip 192.168.10.160 0.0.0.31 192.168.10.0 0.0.0.63
 60 deny ip 192.168.10.160 0.0.0.31 192.168.10.64 0.0.0.63
 70 deny ip 192.168.10.160 0.0.0.31 192.168.10.128 0.0.0.31
 80 deny ip 192.168.10.160 0.0.0.31 192.168.10.192 0.0.0.15
 90 permit ip 192.168.10.160 0.0.0.31 any
Extended IP access list ACL-IT
 10 permit udp any eq bootpc any eq bootps
 20 permit udp any eq bootps any eq bootpc
 30 permit icmp 192.168.10.192 0.0.0.15 192.168.10.0 0.0.0.63
 40 permit icmp 192.168.10.192 0.0.0.15 192.168.10.64 0.0.0.63
 50 permit icmp 192.168.10.192 0.0.0.15 192.168.10.128 0.0.0.31
 60 permit icmp 192.168.10.192 0.0.0.15 192.168.10.160 0.0.0.31
 70 permit icmp 192.168.10.192 0.0.0.15 192.168.10.208 0.0.0.15
 80 permit ip 192.168.10.192 0.0.0.15 192.168.10.0 0.0.0.255
 90 permit ip 192.168.10.192 0.0.0.15 192.168.10.208 0.0.0.15
100 permit ip 192.168.10.192 0.0.0.15 any
```

Justification

The ACL is created on the department server as each department has a specific requirement to who can access them. This provides certain departments with the authorisation to access other departments. In the access-groups, extended ACLs are used since they are more flexible in its implementation. The ip extended access-lists is created for the departments where they permit the appropriate department, allow DHCP and the servers. All the departments restrict IT however, IT has full access to the all the departments. The line ‘permit icmp .. echo-reply’ is where the department responds to the ping while the departments have restricted IT department.

```

PC0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.10.71
Pinging 192.168.10.71 with 32 bytes of data:
Request timed out.
Reply from 192.168.10.71: bytes=32 time=1ms TTL=127
Reply from 192.168.10.71: bytes=32 time=1ms TTL=127
Reply from 192.168.10.71: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.10.71:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms
C:\>ping 192.168.10.132
Pinging 192.168.10.132 with 32 bytes of data:
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Ping statistics for 192.168.10.132:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.10.169
Pinging 192.168.10.169 with 32 bytes of data:
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Ping statistics for 192.168.10.169:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.10.200
Pinging 192.168.10.200 with 32 bytes of data:
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Ping statistics for 192.168.10.200:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.10.227
Pinging 192.168.10.227 with 32 bytes of data:
Request timed out.
Reply from 192.168.10.227: bytes=32 time=1ms TTL=126
Reply from 192.168.10.227: bytes=32 time=1ms TTL=126
Reply from 192.168.10.227: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.10.227:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>

```

Fig 20 – PC in P.M access and restricting appropriately

```

PC7
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.10.7
Pinging 192.168.10.7 with 32 bytes of data:
Reply from 192.168.10.7: bytes=32 time=1ms TTL=127
Reply from 192.168.10.7: bytes=32 time=1ms TTL=127
Reply from 192.168.10.7: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.10.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 2ms
C:\>ping 192.168.10.71
Pinging 192.168.10.71 with 32 bytes of data:
Reply from 192.168.10.71: bytes=32 time=1ms TTL=127
Reply from 192.168.10.71: bytes=32 time=1ms TTL=127
Reply from 192.168.10.71: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.10.71:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.10.132
Pinging 192.168.10.132 with 32 bytes of data:
Request timed out.
Reply from 192.168.10.132: bytes=32 time=1ms TTL=127
Reply from 192.168.10.132: bytes=32 time=1ms TTL=127
Reply from 192.168.10.132: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.10.132:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.10.169
Pinging 192.168.10.169 with 32 bytes of data:
Request timed out.
Reply from 192.168.10.169: bytes=32 time=1ms TTL=127
Reply from 192.168.10.169: bytes=32 time=1ms TTL=127
Reply from 192.168.10.169: bytes=32 time=1ms TTL=127
Ping statistics for 192.168.10.169:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.10.200
Pinging 192.168.10.200 with 32 bytes of data:
Request timed out.
Reply from 192.168.10.200: bytes=32 time=1ms TTL=126
Reply from 192.168.10.200: bytes=32 time=1ms TTL=126
Reply from 192.168.10.200: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.10.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 0ms
C:\>

```

Fig 21– PC in IT accessing department and conference room

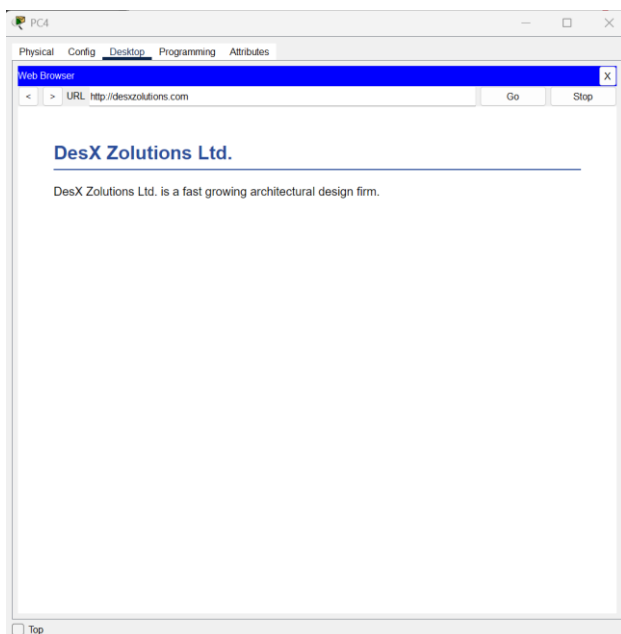


Fig 22- PC in Finance being able to access web page

ZBF

The Zone Based Firewall divides the network into security zones and inspects the traffic between these zones. In the DesX Zolutions Ltd network there are four zones created -

1. Internal – The department router with departments of Design, Project Management, Finance, HR and IT
2. DMZ – This includes the Web, DNS, Mail, DHCP, AAA servers.
3. External – The ISP-Provider
4. Conference – The conference room which is accessible to everyone in the department

```
CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#zone security Internal
CentralRouter(config-sec-zone)#exit
CentralRouter(config)#zone security DMZ
CentralRouter(config-sec-zone)#exit
CentralRouter(config)#zone security External
CentralRouter(config-sec-zone)#exit
CentralRouter(config)#zone security Conference
CentralRouter(config-sec-zone)#exit
CentralRouter(config)#class-map type inspect match-any CP-Internal-Dmz
^
% Invalid input detected at '^' marker.

CentralRouter(config)#class-map type inspect match-any CP-Internal-Dmz
CentralRouter(config-cmap)#match protocol http
CentralRouter(config-cmap)#match protocol https
CentralRouter(config-cmap)#match protocol dns
CentralRouter(config-cmap)#match protocol icmp
CentralRouter(config-cmap)#match protocol smtp
CentralRouter(config-cmap)#match protocol pop3
CentralRouter(config-cmap)#exit
```

After the creation of zones, class maps are created. The match protocol and extended ACL are used to create class maps. For CP-Internal-Dmz, CM-Internal-to-External, CM-Internal-to-Conference, CM-Conference-to-Internal, the match protocol is used which helps to identify the traffic to http, https, icmp and more. This ensures that each zone is allowed the defined set of applications.

```
CentralRouter(config)#ip access-list extended ALC-DMZ-Public
CentralRouter(config-ext-nacl)#permit tcp any host 192.168.10.210 eq 80
CentralRouter(config-ext-nacl)#permit tcp any host 192.168.10.210 eq 443
CentralRouter(config-ext-nacl)#permit tcp any host 192.168.10.212 eq 25
CentralRouter(config-ext-nacl)#permit tcp any host 192.168.10.212 eq 110
CentralRouter(config-ext-nacl)#permit tcp any host 192.168.10.211 eq 53
CentralRouter(config-ext-nacl)#permit udp any host 192.168.10.211 eq 53
CentralRouter(config-ext-nacl)#exit

CentralRouter(config)#IP access-list extended ACL-DHCP-Server-to-Client
CentralRouter(config-ext-nacl)#permit udp any eq 67 any eq 68
CentralRouter(config-ext-nacl)#permit udp any eq 67 any eq 67
CentralRouter(config-ext-nacl)#exit
CentralRouter(config)#end
CentralRouter#
%SYS-5-CONFIG_I: Configured from console by console
```

Fig 23- The access lists is first created which will be mapped into the class-map

```

CentralRouter(config)#class-map type inspect match-any CM-External-to-DMZ-Public

% Invalid input detected at '^' marker.

CentralRouter(config)#class-map type inspect match-any CM-External-to-DMZ-Public
CentralRouter(config-cmap)#match access-group name ALC-DMZ-Public
CentralRouter(config-cmap)#exit
CentralRouter(config)#class-map type inspect match-any CM-Internal-to-DMZ-DHCP
CentralRouter(config-cmap)#match access-group name ACL-DHCP-Client-to-Server
CentralRouter(config-cmap)#exit
CentralRouter(config)#class-map type inspect match-any CM-DMZ-to-Internal-DHCP
CentralRouter(config-cmap)#match access-group name ACL-DHCP-Server-to-Client
CentralRouter(config-cmap)#exit
CentralRouter(config)#class-map type inspect match-any CM-Internal-to-External
CentralRouter(config-cmap)#match protocol http
CentralRouter(config-cmap)#match protocol https
CentralRouter(config-cmap)#match protocol dns
CentralRouter(config-cmap)#match protocol icmp
CentralRouter(config-cmap)#exit
CentralRouter(config)#class-map type inspect match-any CM-Internal-to-Conference
CentralRouter(config-cmap)#match protocol http
CentralRouter(config-cmap)#match protocol https
CentralRouter(config-cmap)#match protocol dns
CentralRouter(config-cmap)#match protocol icmp
CentralRouter(config-cmap)#match protocol smtp
CentralRouter(config-cmap)#match protocol pop3
CentralRouter(config-cmap)#exit
CentralRouter(config)#class-map type inspect match-any CM-Conference-to-Internal
CentralRouter(config-cmap)#match protocol http
CentralRouter(config-cmap)#match protocol https
CentralRouter(config-cmap)#match protocol dns
CentralRouter(config-cmap)#match protocol icmp
CentralRouter(config-cmap)#match protocol smtp
CentralRouter(config-cmap)#match protocol pop3
CentralRouter(config-cmap)#exit
CentralRouter(config)#class-map type inspect match-any CM-Conference-to-DMZ-DHCP
CentralRouter(config-cmap)#match access-group name ACL-DHCP-Client-to-Server
CentralRouter(config-cmap)#exit
CentralRouter(config)#class-map type inspect match-any CM-DMZ-to-Conference-DHCP
CentralRouter(config-cmap)#match access-group name ACL-DHCP-Server-to-Client
CentralRouter(config-cmap)#exit

```

Fig 24 – class map and access-list

```

CentralRouter(config)#class-map type inspect match-any CM-Internal-to-External
CentralRouter(config-cmap)#match protocol http
CentralRouter(config-cmap)#match protocol https
CentralRouter(config-cmap)#match protocol dns
CentralRouter(config-cmap)#match protocol icmp
CentralRouter(config-cmap)#exit
CentralRouter(config)#class-map type inspect match-any CM-Internal-to-Conference
CentralRouter(config-cmap)#match protocol http
CentralRouter(config-cmap)#match protocol https
CentralRouter(config-cmap)#match protocol dns
CentralRouter(config-cmap)#match protocol icmp
CentralRouter(config-cmap)#match protocol smtp
CentralRouter(config-cmap)#match protocol pop3
CentralRouter(config-cmap)#exit
CentralRouter(config)#class-map type inspect match-any CM-Conference-to-Internal
CentralRouter(config-cmap)#match protocol http
CentralRouter(config-cmap)#match protocol https
CentralRouter(config-cmap)#match protocol dns
CentralRouter(config-cmap)#match protocol icmp
CentralRouter(config-cmap)#match protocol smtp
CentralRouter(config-cmap)#match protocol pop3
CentralRouter(config-cmap)#exit

```

After class-maps are created, the appropriate policy-maps are made. For the Internal-to-DMZ policy-map it includes three class maps. The CP-Internal-Dmz matches with https, http, dns and more. The CM-Internal-to-DMZ-DHCP matches with DHCP using ACL and CM-Internal-to-DMZ-RADIUS which matches the RADIUS authentication.

```

CentralRouter(config)#policy-map type inspect PM-Internal-to-DMZ
CentralRouter(config-pmap)#class type inspect CP-Internal-Dmz
CentralRouter(config-pmap-c)#inspect
CentralRouter(config-pmap-c)#exit
CentralRouter(config)#policy-map type inspect CM-Internal-to-DMZ-DHCP
CentralRouter(config-pmap-c)#inspect
%No specific protocol configured in class CM-Internal-to-DMZ-DHCP for inspection. All protocols
will be inspected
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#class class-default
CentralRouter(config-pmap-c)#drop
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#exit
CentralRouter(config)#policy-map type inspect PM-DMZ-to-Internal
CentralRouter(config-pmap-c)#type inspect CM-DMZ-to-Internal-DHCP
CentralRouter(config-pmap-c)#inspect
%No specific protocol configured in class CM-DMZ-to-Internal-DHCP for inspection. All protocols
will be inspected
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#class class-default
CentralRouter(config-pmap-c)#pass
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#policy-map type inspect PM-External-to-DMZ
CentralRouter(config-pmap-c)#class type inspect CM-External-to-DMZ-Public
CentralRouter(config-pmap-c)#inspect
%No specific protocol configured in class CM-External-to-DMZ-Public for inspection. All
protocols will be inspected
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#class class-default
CentralRouter(config-pmap-c)#drop
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#exit
CentralRouter(config)#policy-map type inspect PM-Internal-to-External
CentralRouter(config-pmap-c)#class type inspect CM-Internal-to-External
CentralRouter(config-pmap-c)#inspect
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#class class-default
CentralRouter(config-pmap-c)#drop
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#exit
CentralRouter(config)#policy-map type inspect PM-Internal-to-Conference
CentralRouter(config-pmap-c)#class type inspect CM-Internal-to-Conference
CentralRouter(config-pmap-c)#inspect
% Invalid input detected at '^' marker.

CentralRouter(config-pmap)#class type inspect CM-Internal-to-Conference
CentralRouter(config-pmap-c)#inspect
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#class class-default
CentralRouter(config-pmap-c)#drop
CentralRouter(config-pmap-c)#exit
CentralRouter(config-pmap)#exit
CentralRouter(config)#policy-map type inspect PM-Conference-to-Internal
CentralRouter(config-pmap-c)#class type inspect CM-Conference-to-Internal
CentralRouter(config-pmap-c)#inspect

```

Fig 25 - In policy map, the class maps are added that says what are actions needs to be taken in that class.

The zone pairs links the source zone to the destination zone with the policy maps that were created. Each of the zone-pair defines what traffic is permitted, inspected or dropped. This ensures that all the communication is regulated.

```
CentralRouter#wr
Building configuration...
[OK]
CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#zone-pair security ZP-Internal-to-DMZ source Internal destination DMZ
CentralRouter(config-sec-zone-pair)#service-policy type inspect PM-Internal-to-DMZ
% Invalid input detected at '^' marker.
CentralRouter(config-sec-zone-pair)#service-policy type inspect PM-Internal-to-DMZ
% Invalid input detected at '^' marker.
CentralRouter(config-sec-zone-pair)#service-policy type inspect PM-Internal-to-DMZ
CentralRouter(config-sec-zone-pair)#exit
CentralRouter(config)#zone-pair security ZP-DMZ-to-Internal source DMZ destination Internal
CentralRouter(config-sec-zone-pair)#service-policy type inspect PM-DMZ-to-Internal
% Invalid input detected at '^' marker.
CentralRouter(config-sec-zone-pair)#service-policy type inspect PM-DMZ-to-Internal
CentralRouter(config-sec-zone-pair)#exit
CentralRouter(config)#zone-pair security ZP-External-to-DMZ source External destination DMZ
CentralRouter(config-sec-zone-pair)#service-policy type inspect PM-External-to-DMZ
CentralRouter(config-sec-zone-pair)#exit
CentralRouter(config)#zone-pair security ZP-Internal-to-Conference source Internal destination
Conference
CentralRouter(config-sec-zone-pair)#service-policy type inspect PM-Internal-to-Conference
CentralRouter(config-sec-zone-pair)#exit
CentralRouter(config)#zone-pair security ZP-Conference-to-Internal source Conference destination
Internal
CentralRouter(config-sec-zone-pair)#service-policy type inspect PM-Conference-to-Internal
CentralRouter(config-sec-zone-pair)#exit
```

Fig 21 – The zone-pairs are created, and the policy map is added to the appropriate zone pair

```
CentralRouter#show zone-pair security
Zone-pair name ZP-Internal-to-DMZ
  Source-Zone Internal Destination-Zone DMZ
  service-policy PM-Internal-to-DMZ

Zone-pair name ZP-DMZ-to-Internal
  Source-Zone DMZ Destination-Zone Internal
  service-policy PM-DMZ-to-Internal

Zone-pair name ZP-External-to-DMZ
  Source-Zone External Destination-Zone DMZ
  service-policy PM-External-to-DMZ

Zone-pair name ZP-Internal-to-Conference
  Source-Zone Internal Destination-Zone Conference
  service-policy PM-Internal-to-Conference

Zone-pair name ZP-Conference-to-Internal
  Source-Zone Conference Destination-Zone Internal
  service-policy PM-Conference-to-Internal

Zone-pair name ZP-Conference-to-DMZ-Web
  Source-Zone Conference Destination-Zone DMZ
  service-policy PM-Conference-to-DMZ

CentralRouter#
```

Fig 22 – All the zone-pairs

Lastly, depending on the routers interface, the 'zone-member security' command is placed based on the part of the network it belongs to.

1. The interface that faces towards the company departments are placed in Internal zone.
2. The interface connected to the ServerRouter are placed in the DMZ zone.
3. The interface connected to the ISP-Provider are placed in the External zone.
4. The interface connected to the conference room is assigned the conference zone.

```
%SYS-5-CONFIG-I: Configured from console by console
CentralRouter#wr
Building configuration...
[OK]
CentralRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
CentralRouter(config)#interface s
CentralRouter(config)#interface serial0/1/0
CentralRouter(config-if)#zone-member security Internal
CentralRouter(config-if)#exit
CentralRouter(config)#interface serial0/2/0
CentralRouter(config-if)#zone-member security DMZ
CentralRouter(config-if)#exit
CentralRouter(config)#interface g
CentralRouter(config)#interface gigabitEthernet0/0/0
CentralRouter(config-if)#zone-member security Conference
CentralRouter(config-if)#exit
CentralRouter(config)#interface serial0/1/1
CentralRouter(config-if)#zone-member security External
CentralRouter(config-if)#exit
CentralRouter(config)#end
CentralRouter#
%SYS-5-CONFIG-I: Configured from console by console

CentralRouter#wr
Building configuration...
[OK]
CentralRouter#
```

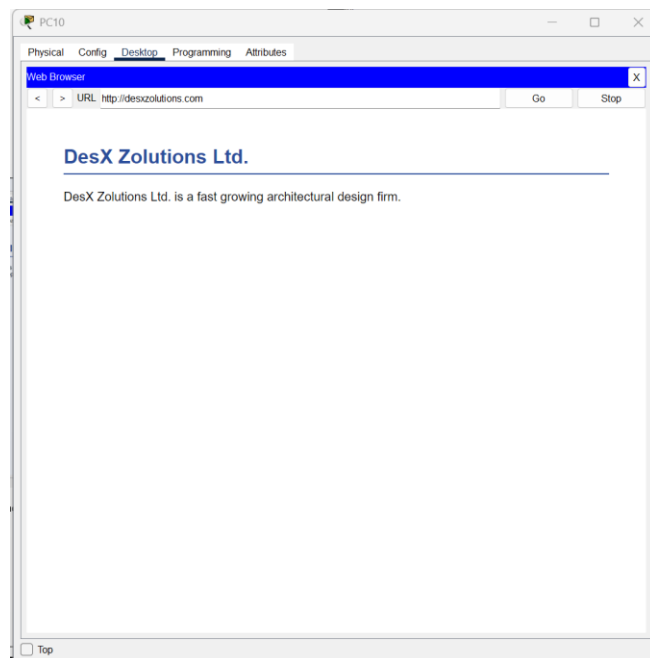
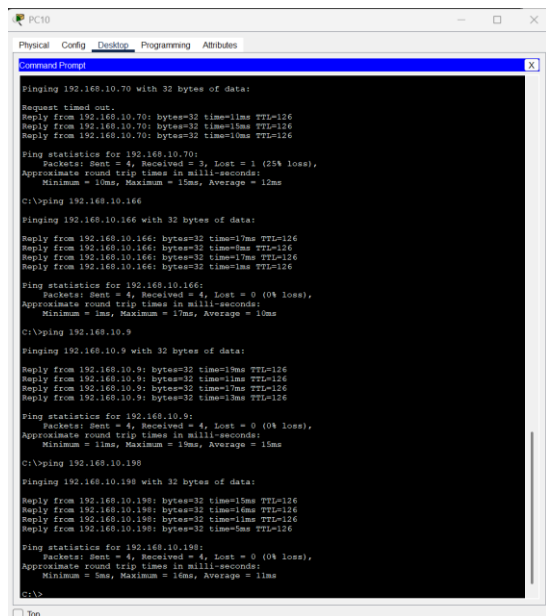
Fig 23 – the zones created and applied to the appropriate interfaces of DepartmentRouter, ISP-Provider, ExternalRouter and ConferenceRoom

Outcome

The Internal users in the Internal zone can access the DMZ services. The AAA also works on the DepartmentRouter without locking one out. The External zone cannot access internal departments. The conference room can access the DMZ [web browser] and the internal department.

```
spanning-tree mode pvst
!
class-map type inspect match-any CP-Internal-Dmz
match protocol http
match protocol https
match protocol dns
match protocol icmp
match protocol smtp
match protocol pop3
match protocol telnet
class-map type inspect match-any CM-External-to-DMZ-Public
match access-group name ACL-DMZ-Public
class-map type inspect match-any CM-Internal-to-DMZ-DHCP
match access-group name ACL-DHCP-Client-to-Server
class-map type inspect match-any CM-DMZ-to-Internal-DHCP
match access-group name ACL-DHCP-Server-to-Client
class-map type inspect match-any CM-Internal-to-External
match protocol http
match protocol https
match protocol dns
match protocol icmp
class-map type inspect match-any CM-Internal-to-Conference
match protocol http
match protocol https
match protocol dns
match protocol icmp
match protocol smtp
match protocol pop3
class-map type inspect match-any CM-Conference-to-Internal
match protocol http
match protocol https
match protocol dns
match protocol icmp
match protocol smtp
match protocol pop3
class-map type inspect match-any CM-Conference-to-DMZ-DHCP
match access-group name ACL-DHCP-Client-to-Server
class-map type inspect match-any CM-DMZ-to-Conference-DHCP
match access-group name ACL-DHCP-Server-to-Client
class-map type inspect match-any CM-Internal-to-DMZ-Radius
match access-group name ACL-Radius-Client
class-map type inspect match-any CM-DMZ-to-Internal-Radius
match access-group name ACL-Radius-Server
class-map type inspect match-any CM-Conference-to-DMZ-Web
match protocol http
match protocol https
match protocol dns
!
policy-map type inspect PM-External-to-DMZ
class type inspect CM-External-to-DMZ-Public
inspect
class type inspect class-default
drop
!
policy-map type inspect PM-Internal-to-External
class type inspect CM-Internal-to-External
inspect
class type inspect class-default
drop
```

Fig 24- ZBF added in the central router seen with 'the show-running config' command



Conference room can ping successfully Project management, Design and IT and also access the webpage.

```

User Access Verification

Password:

ISP-Provider>enable
Password:
ISP-Provider#ping 8.10.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.10.1.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/10/18 ms

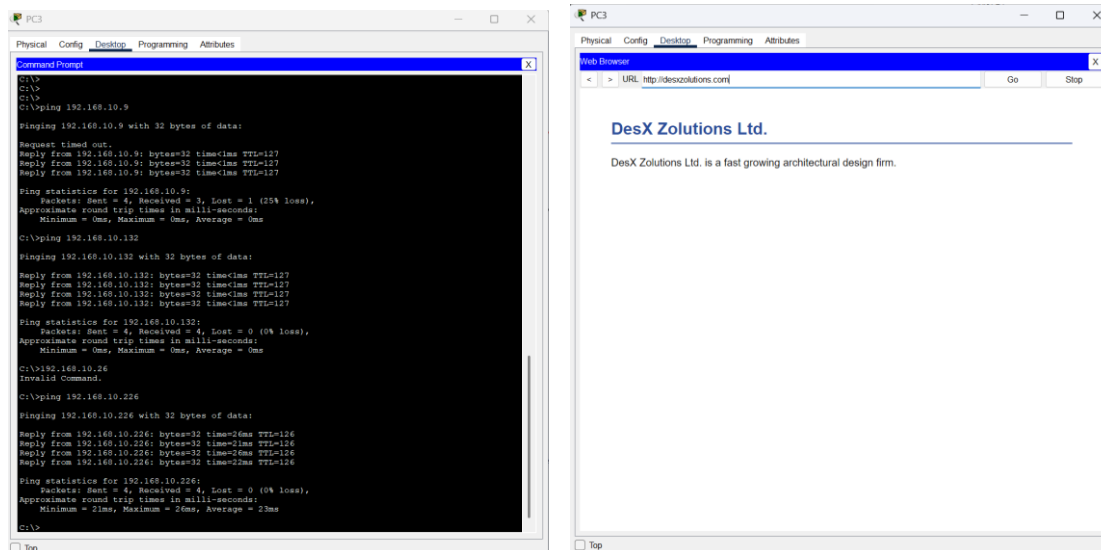
ISP-Provider#ping 192.168.10.7

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.7, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

ISP-Provider#

```

The external ISP-Provider can ping 8.10.1.2 the address of the central router but cannot ping the internal department [design]



The internal network above is the picture of a pc in Project management ping only the departments it has access to according to the department restrictions. It can also host a web page

```

Service-policy inspect : PM-Internal-to-DMZ

Class-map: CP-Internal-Dmz (match-any)
  Match: protocol http
    43 packets, 1766 bytes
    30 second rate 0 bps
  Match: protocol https
    9 packets, 372 bytes
    30 second rate 0 bps
  Match: protocol dns
    9 packets, 545 bytes
    30 second rate 0 bps
  Match: protocol icmp
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol smtp
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol pop3
    0 packets, 0 bytes
    30 second rate 0 bps
  Match: protocol telnet
    3 packets, 132 bytes
    30 second rate 0 bps
  Inspect

Class-map: CM-Internal-to-DMZ-DHCP (match-any)
  Match: access-group name ACL-DHCP-Client-to-Server
    17 packets, 1264 bytes
    30 second rate 0 bps
  Inspect

```

From the dmz zones the devices from internal can access the dmz server and departments are able to get DHCP from the DHCP servers.

Justification

A firewall is to protect the network from any harmful traffic and only allowing the necessary traffic onto the network. There is always a risk for a network to be attacked, internally or externally and it is the important to ensure minimum amount of damage to the company. This can be achieved by segmenting the network into

zones based on its function and sensitivity. The four zones created are Internal, DMZ, Conference and External. The internal department is very critical to the business which needs all the services and require protection from others. The DMZ host public servers like Web, DNS, Email which needs to be reachable to all but also protect critical servers like DHCP and AAA. The external network has untrusted traffic so has minimal access to services. Lastly, the conference room is used and accessed by all users, therefore isolated from the internal departments. All the traffic is passed through the central router which makes it ideal to implement the ZBF. Through ZBF, the policy-maps are applied through the zone-pair, which is assigned to the interfaces. For example, the external network could access the internal department but once the ZBF is set, the user can no longer 'ping' or access other the internal departments.

Task 4: Critical Reflection and Evaluation

The DesX Zolutions Ltd is a simple, straightforward network that allows a growing firm to adapt and make changes to the network design. As the network relies on a few routers, the network design is cost effective. The network is divided into Internal, DMZ, Conference and External zones, which helps to improve the overall network security. All the servers are isolated together in the DMZ which reduces the chances of attacks. However, if the central router fails, then the entire network will crash. The implementation of NAT should also be configured to stimulate internet activities which can help in further external testing. For future growth the network must have backup routers for a large number of users which prevents failure of the entire network. VLAN segmentation can also be implemented which will improve the overall performance and security of the network. The network can also implement VPN which, allows users to access the network remotely. Additionally, protocols like IPSec should also be used to ensure confidentiality integrity and authentication.

In the network, some of the challenges that were faced was in the ACL. The DHCP initially failed until the required UDP ports (67, 68) were added in the extended access-list for the configuration of ACL. The implementation of ZBF was difficult as inaccurate commands of drop and pass were used. This led to the failure of DHCP and AAA authentication. However, by adding the appropriate protocols in class maps and re-ordering the policy maps, the ZBF worked in the end.

Conclusion

The DesX Zolutions is a network design that is configured through OSPF, DHCP, AAA server, ACLs, ZBF and more. This provides a secure and efficient network where departments can have reliable communication. This design also allows for future scalability and improvement making it a flexible solution for a growing firm.